Arithmetic structure of the standard genetic code

Ruslan Khafizov

September 11, 2026

Preprint DOI: 10.5281/zenodo.20751290

Abstract

The standard genetic code is near-universal and stable over evolutionary time, motivating study of its arithmetic properties. We analyze proton and neutron counts of the encoded amino acids (free neutral forms) across 33 codon groups constructed from two natural grouping systems: the Rumer partition by family degeneracy and the three chemical axes of the third codon position (Keto/Amino, Strong/Weak, Purine/Pyrimidine). Excluding the three stop codons and the initiator ATG leaves 60 codons whose group sums determine an arithmetic structure modulo 37. Neutron counts reach the 37-lattice, whereas necessarily even proton counts, barred from odd multiples of 37, fall one below them: globally, and in the single tryptophan/isoleucine pair carrying the Keto/Amino asymmetry, with differences 37 and 36. The Keto/Amino balances and global conditions yield separate neutron and proton equations modulo 37, the former containing $111 = 3 \cdot 37$; since $\gcd(4, 37) = 1$, these equations fix a unique minimal non-negative integer assignment for the four service codons, after which the regularities form a connected family of identities and divisibilities. Across 27 NCBI translation tables, this combination occurs only in the standard code and its exact codon-to-amino-acid equivalent; deterministic controls show divisibilities concentrated in third-position groupings, with 37 standing out among tested prime moduli. Under a random-code null, the observed values of several arithmetic features of this structure fall in the far tails of their Monte Carlo null distributions. Throughout, the regularities are properties of codon-group sums, not individual amino acids, and depend jointly on codon degeneracy and nucleon composition.

Contents

1 Introduction

The genetic code is one of the most ancient and conserved molecular features of cellular organisms. Its near-complete universality across the known domains of life, and the limited scope of variation in the known alternative codes, indicate that the basic structure of the standard code was already established at the early stages of biological evolution. Any organizational regularities present in its arithmetic structure therefore require precise analysis as one of the few potential sources of information about the early stages of life.

The arithmetic structure of the standard genetic code has been studied since shortly after its decipherment. Yu. B. Rumer [6, 7] introduced the partition of the 64 codons into two classes of 32 codons each, Octet I (the eight four-fold degenerate families) and Octet II (the remaining codons), related by the involution $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$. Shcherbak and Makukov [8] observed that the number 37 plays a recurrent role in the arithmetic of the code, appearing as a divisor of amino acid nucleon sums across various groupings of codons. Panov and Filatov [5] revisited this analysis, introducing the per-codon counting rule, which forms the methodological basis of the present study.

The present analysis uses the proton and neutron counts of the twenty canonical amino acids in their free neutral forms under the most-abundant-isotope convention; once these conventions are fixed, the counts are uniquely determined. These quantities are considered together with two natural grouping systems defined independently of the nucleon arithmetic. The first is the Rumer partition. The second is given by the three chemical axes of the third codon position — Keto/Amino, Strong/Weak, and Purine/Pyrimidine — which exhaust the partitions of the four nucleotide bases into two pairs and correspond to the functional-group, hydrogen-bond, and ring-topology dichotomies of nucleobase chemistry.

The present work shows that within this framework the arithmetic regularities of the standard genetic code organize into a connected family of identities and divisibilities. The analysis traces this connected structure to a small number of empirical properties of the standard code. The same arithmetic criteria are then applied to all 27 NCBI translation tables. Among them, the tested combination of structural regularities occurs only in the standard genetic code table and in its exact equivalent with respect to codon-to-amino-acid mapping. The specificity of the structure is examined further by deterministic controls, which test its dependence on the third codon position, on the modulus, and on the molecular representation of the amino acids, and by a Monte Carlo test against an explicit random-code null model.

The analysis is divided into three levels. The principal regularities are established at a parametrization-independent level, at which the three stop codons and ATG in its initiator role are excluded and only the remaining 60 codons are summed; these are the central results of the paper, and they involve no assigned values. To obtain sums for the complete table of 64 codons, however, the four service codons must be given some nucleon contributions. The second level is the naive reading (Key 0), in which the stop codons are assigned zero and ATG is counted as methionine. Assigning zero is also a choice of value, not the absence of one. The third level is the symmetrization (Key 1), in which the four service codons carry derived nucleon contributions. Key 1 does not establish the parametrization-independent core of the structure: that core is already present in sums over the 60 codons remaining after the service codons are excluded. The role of Key 1 is to show how the structure extends to the complete table, taking into account the fixed distribution of the service codons across the groups in the standard code. For the chosen symmetry conditions, these contributions are determined as the unique minimal non-negative integer solution of the corresponding system (Section 3), and the resulting regularities are consequences of that solution.

Assigning values to the service codons is a formal device for treating the arithmetic of the complete table as a single object; neither the assigned values nor the sums that include them are given a physical interpretation.

2 Methods

This section defines the nucleon parameters, the codon group systematics, and the parametrizations of service codons used throughout the analysis, together with the computed datasets supporting it.

2.1 Nucleon composition of amino acids

Throughout this work, each of the twenty canonical amino acids is represented as a free neutral amino acid molecule (not as a residue in a peptide chain), with its molecular formula retrieved from the PubChem compound database [3]. From this formula, we compute four integer quantities that characterize the nucleon content of the molecule. Below, $Q(a)$ denotes any of the four quantities T, P, N, or Δ for one molecule of amino acid a. The sensitivity of the results to this choice of molecular representation is examined in Section 4.2.

The proton count P is defined from the elemental composition of the molecular formula as

$$P = \sum_E n_E Z_E,$$

where n_E is the number of atoms of element E in the formula and Z_E is its atomic number. The atomic numbers used are $Z(\mathrm{H}) = 1$, $Z(\mathrm{C}) = 6$, $Z(\mathrm{N}) = 7$, $Z(\mathrm{O}) = 8$ and $Z(\mathrm{S}) = 16$.

The neutron count N is computed by assigning each element its most abundant naturally occurring stable isotope, using the terrestrial isotopic compositions evaluated by IUPAC [1].

Accordingly, we use ^{1}H, ^{12}C, ^{14}N, ^{16}O and ^{32}S. The total neutron count is

$$N = \sum_E n_E(A_E - Z_E),$$

where A_E is the mass number of the selected isotope of element E.

The total nucleon count T and the proton–neutron difference Δ are defined as

$$T = P + N, \qquad \Delta = P - N.$$

Both quantities are integers by construction. Under the isotope assumptions above, hydrogen is the only element contributing to Δ, since all other isotopes used have equal numbers of protons and neutrons. Consequently, Δ is numerically equal to the number of hydrogen atoms in the molecule and, in particular, $\Delta \geq 0$ for each of the twenty canonical amino acids. By linearity, the same identity holds for any sum of canonical amino-acid contributions and, in particular, for all group sums over the service-excluded pool (Section 2.2).

To avoid a notational collision, we use uppercase Δ exclusively for the proton–neutron imbalance of a single molecule or codon group, and reserve lowercase δ for pairwise differences between the molecular quantities of two amino acids (for example, $\delta P = P_i - P_j$ between two amino acids i and j). When both amino acids are identified explicitly, their labels are placed in parentheses:

$$\delta Q(a, b) = Q(a) - Q(b).$$

We emphasize that P, N, T and Δ are not measured molecular masses. They are exact integer quantities derived from elemental composition, and all arithmetic identities reported in this study are exact integer relations rather than artifacts of rounding approximate molecular masses.

The nucleon values for all twenty canonical amino acids are collected in the input dataset `amino_acids_nucleons.csv`, with one row per amino acid and columns for P, N, T and Δ. From this table we compute three companion datasets of pairwise absolute differences: `amino_acids_proton_differences.csv`, `amino_acids_neutron_differences.csv` and `amino_acids_nucleon`
For any pair of amino acids (i, j) we write $\delta P = P_i - P_j$ and analogously for N and T. Each file groups all $\binom{20}{2} = 190$ unordered pairs (i, j) by the corresponding absolute difference $|\delta P(i, j)|$, $|\delta N(i, j)|$, or $|\delta T(i, j)|$.

All subsequent nucleon computations for codon groups reduce, by linearity, to sums of the per-amino-acid values tabulated in `amino_acids_nucleons.csv`, weighted by the multiplicities of the corresponding codons. Where service codons are included, their contributions are determined by the chosen parametrization (Section 2.5).

2.2 Genetic code structure

The standard genetic code maps the 64 codons of the four-letter nucleotide alphabet $\{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}$ onto the twenty canonical amino acids and a translation-termination signal. Throughout this paper, we use DNA notation (with thymine T rather than uracil U) and adopt the codon assignments of The Standard Code (NCBI translation table 1) [4]. The three codons TAA, TAG and TGA act as stop signals, whereas ATG encodes methionine and serves as the canonical translation-initiation signal. For the purposes of the present arithmetic analysis, these four codons — the three stop codons and ATG in its initiator role — are referred to as *service codons*. Among possible initiation codons, ATG is predominant. Its RNA equivalent AUG is the only one of them fully complementary to the CAU anticodon of the initiator methionine tRNA. In the present analysis, ATG represents the initiation function; this analytical convention does not exclude initiation from other codons.

Let the complete set of 64 codons be

$$\mathcal{C}_{64} = \{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}^3,$$

and let the four service codons form the set

$$\mathcal{S} = \{\text{ATG}, \text{TAA}, \text{TAG}, \text{TGA}\}.$$

The *service-codon-excluded 60-codon pool* is then defined as

$$\mathcal{C}_{\text{SE}} = \mathcal{C}_{64} \setminus \mathcal{S},$$

with $|\mathcal{C}_{\text{SE}}| = 60$. Every codon belonging to $\mathcal{C}_{\text{SE}}$ is counted according to its standard amino-acid assignment. Because ATG is the sole methionine codon, $\mathcal{C}_{\text{SE}}$ contains codons for nineteen canonical amino acids; methionine is absent from this pool. The full codon-to-product mapping is provided in the dataset `genetic_code_codons.csv`.

By a codon family, we mean the set of four codons sharing the same first two bases and having an arbitrary third-position base, denoted by N. A structural property of the standard code, first identified by Rumer [6, 7], is the partition of the 64 codons into two halves of 32 codons each, distinguished by the degeneracy pattern of the third codon position. The first half, which we denote *Octet I*, consists of the eight codon families in which all four third-position nucleotides encode the same amino acid. These fully four-fold degenerate families are GCN (Ala), CGN (Arg), GGN (Gly), CCN (Pro), ACN (Thr), GTN (Val), CTN (Leu) and TCN (Ser). The second half, denoted *Octet II*, consists of the remaining eight codon families, in which the third-position nucleotide affects the encoded product. Every service codon of the standard code belongs to Octet II and carries a purine (A or G) at the third position.

The Rumer partition is not an analytical construct introduced in the present work but a structural feature of the code itself. It reflects both the qualitative distinction between fully and partially degenerate codon families and the quantitative fact that these two classes have equal cardinality: each contains eight families and thirty-two codons. Because Octet I contains no service codons, all nucleon quantities computed over Octet I are independent of any parametrization of start and stop signals and are determined solely by the nucleon content of the encoded amino acids. The same property holds, for a separate reason, for any codon group whose third-position nucleotide is restricted to pyrimidines (cytosine or thymine), since all four service codons of the standard code carry a purine at the third position. Full four-fold degeneracy and a purine-free third position are two distinct structural conditions, but both have the same consequence: nucleon quantities over the corresponding groups are fixed by their amino-acid composition and provide parametrization-independent reference values for the arithmetic identities of Section 3.

A further structural property of the Rumer partition, introduced by Rumer [7], is that the two octets are related by a simple permutation of the nucleotide alphabet. Under the transformation

$$R = (\text{T} \leftrightarrow \text{G},\ \text{C} \leftrightarrow \text{A}),$$

each codon family of Octet I is mapped to a codon family of Octet II, and vice versa. This transformation, known as the Rumer transformation, rearranges the partition but does not preserve the amino acids encoded by individual codons.

Taken together, the four-fold degeneracy pattern and the Rumer transformation identify the Octet I / Octet II partition as a natural structural division of the standard code, independent of the nucleon arithmetic. In the present work, this division is used as a reference partition over which nucleon quantities are computed and compared.

The codon table (Figure 1) is displayed with its axes in the base ordering C, G, T, A, which is also the default ordering of the accompanying visualizer. This ordering has two justifications. Chemically, it was introduced by Rumer [7] on the basis of base type and the number of hydrogen bonds in the canonical base pair. The sequence C, G, T, A alternates pyrimidines and purines, while within each of these classes the member of the Strong pair $\{\text{C}, \text{G}\}$ precedes the member of the Weak pair $\{\text{T}, \text{A}\}$: C precedes T among the pyrimidines, and G precedes A among the purines. Geometrically, as shown by Panov and Filatov [5], it is one of exactly four orderings (out of 24 possible permutations of the four bases) in which Octet I and Octet II form connected regions on the codon matrix.

To assess the extent to which the arithmetic properties studied in this paper are specific to the standard genetic code, we additionally retrieved the full set of 27 translation tables maintained by NCBI [4]. The tables are numbered 1–33, with some numbers left unused, and cover nuclear, mitochondrial and plastid codes across a broad taxonomic range. The codon assignments of all these tables are collected in the dataset `ncbi_genetic_code_registry.csv` and are used in Section 4.1 to compare the standard code with the other NCBI translation tables.

	C					G					T					A				
C	CCC	Pro	115	62	53	CGC	Arg	174	94	80	CTC	Leu	131	72	59	CAC	His	155	82	73
	CCG	Pro	115	62	53	CGG	Arg	174	94	80	CTG	Leu	131	72	59	CAG	Gln	146	78	68
	CCT	Pro	115	62	53	CGT	Arg	174	94	80	CTT	Leu	131	72	59	CAT	His	155	82	73
	CCA	Pro	115	62	53	CGA	Arg	174	94	80	CTA	Leu	131	72	59	CAA	Gln	146	78	68
G	GCC	Ala	89	48	41	GGC	Gly	75	40	35	GTC	Val	117	64	53	GAC	Asp	133	70	63
	GCG	Ala	89	48	41	GGG	Gly	75	40	35	GTG	Val	117	64	53	GAG	Glu	147	78	69
	GCT	Ala	89	48	41	GGT	Gly	75	40	35	GTT	Val	117	64	53	GAT	Asp	133	70	63
	GCA	Ala	89	48	41	GGA	Gly	75	40	35	GTA	Val	117	64	53	GAA	Glu	147	78	69
T	TCC	Ser	105	56	49	TGC	Cys	121	64	57	TTC	Phe	165	88	77	TAC	Tyr	181	96	85
	TCG	Ser	105	56	49	TGG	Trp	204	108	96	TTG	Leu	131	72	59	TAG	STOP	0	0	0
	TCT	Ser	105	56	49	TGT	Cys	121	64	57	TTT	Phe	165	88	77	TAT	Tyr	181	96	85
	TCA	Ser	105	56	49	TGA	STOP	0	0	0	TTA	Leu	131	72	59	TAA	STOP	0	0	0
A	ACC	Thr	119	64	55	AGC	Ser	105	56	49	ATC	Ile	131	72	59	AAC	Asn	132	70	62
	ACG	Thr	119	64	55	AGG	Arg	174	94	80	ATG	Met	149	80	69	AAG	Lys	146	80	66
	ACT	Thr	119	64	55	AGT	Ser	105	56	49	ATT	Ile	131	72	59	AAT	Asn	132	70	62
	ACA	Thr	119	64	55	AGA	Arg	174	94	80	ATA	Ile	131	72	59	AAA	Lys	146	80	66

Figure 1: Standard genetic code with nucleon parameters of the encoded amino acids, displayed under the naive reading (Key 0), in which stop codons contribute zero and ATG is counted as methionine. In each cell, the codon appears on the left; the symbol of the encoded product (the three-letter amino acid symbol or STOP for the three stop codons) appears at the top center; the total nucleon count T appears below the product symbol; the proton count P appears at the top right; and the neutron count N appears at the bottom right. Codons of Octet I (the eight fully four-fold degenerate families) are shaded purple, and codons of Octet II are shaded yellow. Generated by the accompanying `visualization.html`.

2.3 Codon-group construction

In addition to the Rumer partition introduced in Section 2.2, we use an additional overlapping grouping system based on the chemical identity of the nucleotide at the third codon position.

Each of the four nucleotides belongs to one of two classes under each of three distinct binary chemical classifications:

- base type: *purine* $\{\mathrm{A}, \mathrm{G}\}$ (R) or *pyrimidine* $\{\mathrm{C}, \mathrm{T}\}$ (Y);
- number of hydrogen bonds in the canonical base pair: *strong* $\{\mathrm{C}, \mathrm{G}\}$ (S; three bonds) or *weak* $\{\mathrm{A}, \mathrm{T}\}$ (W; two bonds);
- functional class: *amino* $\{\mathrm{A}, \mathrm{C}\}$ (M) or *keto* $\{\mathrm{G}, \mathrm{T}\}$ (K).

These axes were not selected in response to the subsequent nucleon calculations: they are three conventional pairwise dichotomies of the nucleotides and are represented explicitly in the IUPAC–IUB ambiguity nomenclature by the symbol pairs R/Y, S/W, and M/K. Together they exhaust the three possible partitions of four nucleotides into two unordered pairs, thereby yielding six two-element subsets $X \subset \{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}$.

For each of these six two-element subsets, and for each singleton $X \in \{\{\mathrm{A}\}, \{\mathrm{C}\}, \{\mathrm{G}\}, \{\mathrm{T}\}\}$, we define the codon group

$$G_X = \{(c_1, c_2, c_3) \in \mathcal{C}_{64} : c_3 \in X\}.$$

Thus, G_X consists of all codons whose third-position nucleotide belongs to X. The six two-nucleotide groups overlap: every codon belongs to exactly three of them, one for each chemical classification. At the same time, every codon belongs to exactly one of the four one-nucleotide groups.

This grouping system is applied at three levels: to all 64 codons, to the 32 codons of Octet I, and to the 32 codons of Octet II. If $|X| = 2$, then G_X contains 32 codons, while each of the intersections $G_X \cap \text{Octet I}$ and $G_X \cap \text{Octet II}$ contains 16 codons. If $|X| = 1$, the corresponding sizes are 16 and 8 codons. Group size alone does not identify a particular group; formulas therefore specify the set X and, where necessary, the octet explicitly.

The family of 33 codon groups used in the present work comprises the full code as one group of 64 codons; the two octets as groups of 32 codons each; ten groups G_X at the full-code level, corresponding to the six two-nucleotide and four one-nucleotide sets X; and their twenty intersections with Octet I and Octet II. Thus, the total number of analyzed groups is

$$1 + 2 + 10 + 20 = 33.$$

For example, $G_{\{\mathrm{G},\mathrm{T}\}} \cap \text{Octet I}$ is the 16-codon group $\text{Keto} \cap \text{Octet I}$, whereas $G_{\{\mathrm{A}\}} \cap \text{Octet II}$ is the 8-codon group of Octet II with adenine at the third position. All 33 groups, together with their sizes, codon compositions, and Rumer-octet membership, are listed in the dataset `codon_groups.csv`.

In the group registry, the full set $\mathcal{C}_{64}$ corresponds to the row with id 1. In this article, that group is denoted All, whereas $\mathcal{C}_{64}$ denotes the codon set itself.

2.4 Pyrimidine synonymy and twin-group identities

Within every codon family of the standard genetic code, the codons NNC and NNT with identical first two positions encode the same amino acid. Therefore, exchanging $\mathrm{C} \leftrightarrow \mathrm{T}$ at the third position does not change the multiset of encoded amino acids. This exchange maps Keto $\{\mathrm{G}, \mathrm{T}\}$ to Strong $\{\mathrm{C}, \mathrm{G}\}$, and Amino $\{\mathrm{A}, \mathrm{C}\}$ to Weak $\{\mathrm{A}, \mathrm{T}\}$.

The two members of each pair contain the same service codons: Keto and Strong contain ATG and TAG, whereas Amino and Weak contain TAA and TGA. Therefore, under any assignment K that associates a pair of values (P, N) with each service codon (Section 2.5) and for $Q \in \{T, P, N, \Delta\}$, the identities

$$Q_K(\text{Keto}) = Q_K(\text{Strong}), \qquad Q_K(\text{Amino}) = Q_K(\text{Weak})$$

hold. Intersecting both groups in either pair with the same octet preserves these identities. The equalities are independent of parametrization, although the full-group values themselves may depend on K. We refer to the Keto/Strong and Amino/Weak pairs, and to the corresponding pairs formed by intersection with the same octet, as *twin groups*. In the dataset `codon_groups.csv`, this relationship is recorded in the `Twin_Group_ID` column.

Purine and Pyrimidine are not twin groups: exchanging C $\leftrightarrow$ T preserves each of these groups rather than mapping one to the other. The same pyrimidine synonymy directly gives equality of all four nucleon quantities for the one-nucleotide groups {C} and {T} and for their intersections with either octet. These groups and their intersections with the octets contain no service codons, so both the values themselves and the equalities between them are independent of parametrization. Because this is a pair of one-nucleotide groups rather than a pair of two-nucleotide branches of the chemical axes, the term *twin groups* and the `Twin_Group_ID` column are not used for it.

2.5 Parametrization of service codons

The nucleon quantities of a codon group are obtained by summing the values of the amino acids encoded by its constituent codons in the standard genetic code. For codons counted only according to their amino-acid assignments, this summation is unambiguous. A separate convention is needed for the three stop codons, which have no amino-acid product, and for ATG in its initiator role; ATG also encodes methionine (Section 2.2).

The complete codon set $\mathcal{C}_{64}$, the four-codon service set $\mathcal{S}$, and the remaining 60-codon pool $\mathcal{C}_{\mathrm{SE}}$ are defined in Section 2.2. We refer to $\mathcal{C}_{\mathrm{SE}}$ below as the *service-excluded pool*; its technical identifier in the code and file names is `pool60`. Every codon in $\mathcal{C}_{\mathrm{SE}}$ is unambiguously assigned to a canonical amino acid. For $c \in \mathcal{C}_{\mathrm{SE}}$ and $Q \in \{T, P, N, \Delta\}$, $Q(c)$ denotes the corresponding quantity of the free neutral form of the amino acid encoded by c, as defined in Section 2.1.

For any codon set $A \subseteq \mathcal{C}_{64}$ containing no service codons, define

$$Q(A) := \sum_{c \in A} Q(c).$$

For a codon group G, its sum over the service-excluded pool is defined by

$$Q_{\mathrm{SE}}(G) := \sum_{c \in G \cap \mathcal{C}_{\mathrm{SE}}} Q(c).$$

For a service codon c, the notation $Q(c)$ without a key subscript is not used because the formal contribution of such a codon depends on the selected key. The subscript SE indicates a restriction of the summation domain, not a parametrization key. Neither the sum $Q_{\mathrm{SE}}(G)$ nor the number of codons included in it, $|G \cap \mathcal{C}_{\mathrm{SE}}|$, depends on the assignments to the service codons.

If the group G itself contains no service codons, the restriction removes nothing and

$$Q(G) = Q_{\mathrm{SE}}(G).$$

To analyze the complete 64-position table, we introduce a *parametric key* K: a fixed assignment of a pair of numbers $\big(P_K(c), N_K(c)\big)$ to every service codon c. Under such a key, the group sums are defined by

$$\big(P_K(G), N_K(G)\big) := \big(P_{\mathrm{SE}}(G), N_{\mathrm{SE}}(G)\big) + \sum_{c \in G \cap \mathcal{S}} \big(P_K(c), N_K(c)\big),$$

and the remaining two quantities are derived rather than assigned independently:

$$T_K(G) = P_K(G) + N_K(G), \qquad \Delta_K(G) = P_K(G) - N_K(G).$$

The same two relations define $T_K(c)$ and $\Delta_K(c)$ for each service codon c. For $Q \in \{T, P, N, \Delta\}$, $Q_0(G)$ and $Q_1(G)$ below denote the full group sums, including service-codon contributions, under the Key 0 and Key 1 assignments, respectively. The subscripts 0 and 1 label these two assignments; they are not numerical values of a scalar parameter. Thus $Q_{\mathrm{SE}}(G)$ is independent of the key, whereas $Q_0(G)$ and $Q_1(G)$ can differ only through the contributions of the service codons contained in group G. If group G contains no service codons, then $Q_K(G) = Q(G) = Q_{\mathrm{SE}}(G)$ for every K. The subscript SE applies only to codon groups; the subscripts K, 0, and 1 apply to full group sums and to the formal contributions of individual service codons. The two keys considered in this study are recorded in `key_parameters.csv`.

The naive reading (Key 0). Each of the three stop codons is assigned the zero contribution $(P, N) = (0, 0)$ because a termination signal has no amino acid product. The ATG codon is treated as an ordinary methionine codon and receives the values of the free methionine molecule, $(P, N) = (80, 69)$. This simple convention corresponds to the ordinary reading of the amino-acid-coding part of the table, in which the stop signals are assigned a zero contribution, but it remains a parametrization. In particular, Key 0 is not identical to the service-excluded pool: Key 0 retains all 64 positions and counts ATG as methionine, whereas in the pool the summation runs over the remaining 60 codons only.

Symmetrization (Key 1). Each of the three stop codons is assigned $(P, N) = (37, 37)$, while ATG, treated as a formal initiation marker, is assigned $(P, N) = (1, 0)$. These values are not interpreted as the nucleon composition of physical translation products. They are derived in Sections 3.4–3.4.3 under the assumption that the three stop codons share a common parameter pair and under the conditions of balance and divisibility by 37 specified there. Under these assumptions, Key 1 is the unique minimal non-negative integer solution of the resulting system. Minimality here means the smallest total contribution of the service codons—the sum of the contributions of the three stop codons and ATG—determined separately for protons and for neutrons. For the standard code the balance fixes the difference between the stop and start parameters, so the total contribution increases with the start parameter and is minimal at its smallest admissible value. Here Key 1 is introduced as the result of that formal construction, not as a biochemical property of termination signals or the initiator.

For codon groups containing no service codons, the values under Key 0 and Key 1 coincide with the direct amino-acid sums. If a group contains service codons, the Key 1 assignments have only a formal arithmetic meaning: the corresponding sums do not represent the nucleon composition of a physical ensemble of molecules. The terms "proton count", "neutron count" and "total nucleon count" are retained for consistency with the underlying amino acid contributions, but no biochemical interpretation is claimed for the formal Key 1 contributions.

The dataset `key_parameters.csv` additionally contains eight reference assignments, `Key 2`–`Key 9`, considered during the exploratory stage. They are not used to obtain the results of this study and do not have the status of the derived Key 1. The following analysis therefore distinguishes three levels explicitly: the service-excluded pool, the naive reading (Key 0), and the symmetrization (Key 1).

2.6 Analytical criteria and classification of equalities

For each of the 33 codon groups defined in Section 2.3, we examine P, N, T, and Δ at three analytical levels: the service-excluded pool, Key 0, and Key 1. Three reporting procedures are described below: tests of divisibility by 37 and decimal roundness, and a search for exact equalities. Divisibility by 37 is the primary preselected arithmetic criterion. Exact powers, reduced ratios, and matches to amino acid differences are considered separately as additional descriptive observations. Statistical significance is assessed only where a corresponding null model is specified.

Criterion 1: divisibility by 37. For every group quantity P, N, T and Δ, we compute the quotient and remainder upon division by 37. A zero remainder denotes exact divisibility. This criterion continues the line of investigation opened by Shcherbak and Makukov [8] and developed by Panov and Filatov [5]. When a value is divisible by 37, its displayed factorization also exposes any further factors of the quotient; these are not assigned separate evidential weight without an additional analysis. Nonzero residues are reported descriptively only when they enter exact relations among predefined groups or form recurring patterns of one-unit deviations from the nearest multiple of 37.

Complete expressions of the form $q \cdot 37 + r$ for all groups and all four quantities are provided in `pool60_divisibility-37.csv`, `key0_divisibility-37.csv` and `key1_divisibility-37.csv`.

Criterion 2: roundness in the decimal system. We call a positive integer quantity round in the decimal system if it is divisible by 10^k for at least one integer $k \geq 1$; its degree of roundness is the largest such k. This criterion is used descriptively because the data contain exact multiples of 100 and 1000, notably among the parametrization-independent Octet I sums. Although roundness is tied to decimal notation, the divisibility being tested is a property of the integer itself: $10^k \mid x$ is equivalent to simultaneous divisibility by 2^k and 5^k. The systematic control across prime moduli is presented in Section 4.2.

Criterion 3: exact equalities. At each analytical level, we compare the P, N, T, and Δ values of all 33 groups and record every exact match. Trivial identities such as $P(G) = P(G)$ are not included in the results. Each equality is assigned to one of three classes:

- *different-size equalities* (`DIFF_PARAM` in the `Match_Type` column): the same numerical value is obtained by summing over different numbers of codons; either the same or different nucleon quantities may be compared;
- *same-size cross-quantity equalities* (`CROSS_PARAM`): the groups contain the same number of codons, but different quantities are compared, for example $P(G_1) = N(G_2)$;
- *same-size same-quantity equalities* (`SAME_PARAM`): the groups contain the same number of codons and the same quantity is compared, for example $P(G_1) = P(G_2)$.

For classification, group size is the number of codons actually included in the sum. Under Key 0 and Key 1, this is the full size of the group in the 64-codon table. In the service-excluded pool, the service codons are removed, so groups containing them are smaller.

All identified equalities are listed in `pool60_equalities.csv`, `key0_equalities.csv` and `key1_equalities.csv`. Twin-group identities from Section 2.4 are also recorded, but are treated as consequences of identical amino acid composition rather than as new coincidences.

Additional exploratory observations. Beyond the primary procedures, we descriptively note:

- exact powers of small integers, for example $4096 = 2^{12}$;
- reduced ratios within a group, including P/T, N/T, Δ/T, P/N, Δ/P and Δ/N, tabulated in `pool60_ratios.csv`, `key0_ratios.csv` and `key1_ratios.csv`;
- matches between group-level quantities and the absolute differences $|\delta P|$, $|\delta N|$ or $|\delta T|$ of individual amino acid pairs, such as $|\delta N(\mathrm{Trp}, \mathrm{Ile})| = 37$.

These observations describe the arithmetic and may provide a basis for testable hypotheses.

Finally, the four quantities are not independent:

$$T = P + N, \qquad \Delta = P - N.$$

Equalities, divisibility relations or ratios obtained from one another through these identities are therefore not counted as separate evidence. The same applies to copies propagated through nested groups or twin-group identities. Section 3 distinguishes (i) facts computed directly for the service-excluded pool without assigning values to the service codons; (ii) identities whose invariance to the key is established algebraically; and (iii) results specific to Key 0 or Key 1.

2.7 Computed datasets and reproducibility

The principal deterministic results of this study are produced by `reproduce.py`. The script reads five input datasets and writes 17 computed output datasets. A separate deterministic script, `controls.py`, writes four control datasets, whereas the statistical analysis is performed by `mc_significance.py`, which writes the summary table `mc_significance.csv`, and is independently re-checked by `verify_mc.py`. By default, all data paths are resolved relative to the directory of the corresponding script.

Input datasets. Five input files define the numerical and structural basis of the analysis:

- `amino_acids_nucleons.csv` contains P, N, T and Δ for the twenty canonical amino acids (Section 2.1);
- `genetic_code_codons.csv` specifies the standard codon-to-translation-product mapping (Section 2.2);
- `ncbi_genetic_code_registry.csv` records a snapshot of the 27 NCBI translation tables, including the standard table (Section 2.2);
- `codon_groups.csv` defines the system of 33 overlapping codon groups (Section 2.3);
- `key_parameters.csv` contains the service-codon assignments of the parametric keys (Section 2.5).

Per-key computed datasets. For each of the two principal keys, Key 0 and Key 1, `reproduce.py` writes four files. The prefixes `key0_` and `key1_` identify the active key:

- `*_nucleon-data.csv` contains the four reported group quantities P, N, T and Δ for all 33 groups; T and Δ are derived from P and N;
- `*_divisibility-37.csv` expresses every quantity exactly as $q \cdot 37 + r$;
- `*_equalities.csv` exhaustively enumerates exact numerical equalities and classifies them in the `Match_Type` column as `DIFF_PARAM`, `CROSS_PARAM` or `SAME_PARAM`, following Section 2.6;
- `*_ratios.csv` contains the six ratios P/T, N/T, Δ/T, P/N, Δ/P and Δ/N both as exact reduced fractions and as rounded decimal approximations.

The two keys therefore produce eight output files.

Service-excluded-pool datasets. For the service-excluded pool, the script writes four analogous files: `pool60_nucleon-data.csv`, `pool60_divisibility-37.csv`, `pool60_equalities.csv` and `pool60_ratios.csv`. Before summation, the service codons are removed from every group G, so that the number of codons included is $|G \cap \mathcal{C}_{\mathrm{SE}}|$. No parametric key is applied.

Amino acid differences. Three files written once, `amino_acids_proton_differences.csv`, `amino_acids_neutron_differences.csv` and `amino_acids_nucleon_differences.csv`, enumerate the absolute differences $|\delta P|$, $|\delta N|$ and $|\delta T|$, respectively, for all $\binom{20}{2} = 190$ unordered pairs of canonical amino acids.

Table-dependent NCBI pools. The remaining two outputs of the core script compare the standard code with the other tables in `ncbi_genetic_code_registry.csv`. Because the sets of stop codons differ between tables, we do not assume in advance that every analyzed pool contains 60 codons. Two models are used. Model 1 excludes ATG and every codon listed in the `Stop_Codons` field. Model 2 excludes ATG and only the permanent stop codons marked `*` in the `Amino_Acids` field; context-dependent stops are included in the analysis with the amino acid assignments recorded for them. The models coincide for tables without context-dependent stops. For the standard table, both models give the same 60-codon service-excluded pool. The actual number of codons in the analyzed pool is recorded for both models (columns `Pure_Codons_Count` and `Pure_Codons_Count_Alt`).

`deficit_models_analysis.csv` contains one row for each of the 27 tables. The full proton and neutron sums follow the registered amino acid assignments, with positions marked `*` contributing zero. For both models, the file additionally records sums over the corresponding analyzed pool, the proton-sum residue modulo 37, and three neutron deficits relative to:

1. the constant 3700 obtained for standard-code Octet I;
2. structural T(Octet I), retaining the eight families that occupy Octet I positions in the standard table;
3. functional T(Octet I), selecting in table t those four-codon families that have one amino acid assignment and no permanent stop codon.

`keto_amino_balance_models.csv` uses the same two models and, for every table, sums P and N separately over the intersections of the analyzed pool with the third-position Keto and Amino groups. Both sums and the directed Keto-minus-Amino difference are recorded.

Reproducibility of the core pipeline. `reproduce.py` uses only the Python standard library and requires neither external packages nor a virtual environment. The computation is deterministic: outputs are written in UTF-8 with LF line endings and a fixed row order. The canonical repository files are regenerated from the five inputs without manual editing. All group sums and residues use integer arithmetic, while ratios are first reduced exactly by their greatest common divisor. The decimal columns in `*_ratios.csv` are rounded representations of exact fractions and are not used to obtain any result. Every deterministic numerical claim can be traced to a specific cell or to a documented exact combination of cells in the output files.

Neither `reproduce.py` nor `controls.py` uses randomization. Randomization is confined to the separate Monte Carlo analysis in `mc_significance.py`, with a fixed seed and a specified number of trials. Its tail-probability estimates and uncertainty intervals necessarily use floating-point values. `verify_mc.py` independently re-implements the same analysis; the statistical procedure is fully specified in Section 4.3.

Supplementary visualizer. `visualization.html` provides an interactive view of the translation tables and codon groups under the service-excluded pool (POOL60), Key 0 and Key 1, together with additional exploratory views. It is not part of the core reproducible pipeline and is not the source of published numerical results: values reported in the main text are checked against the deterministic CSV files. An online version is available at `https://ruslankhafizov.github.io/genetic-code-arithmetic/`.

Supplementary control datasets. `controls.py` reads the same five input files, uses only the Python standard library and writes four deterministic datasets:

- `position_axis_analysis.csv` separately applies the three binary chemical classifications Keto/Amino, Strong/Weak and Purine/Pyrimidine to the first, second and third codon positions at three analytical levels: the service-excluded pool, Key 0 and Key 1. For each side it records the four nucleon quantities and their residues modulo 37;

- `prime_divisibility_scan.csv` scans every prime up to 97 over three value sets: group quantities identical under Key 0 and Key 1, all Key 0 quantities, and all Key 1 quantities. For each prime it records the numbers of divisible instances, distinct nonzero values and distinct odd cores, together with a reference value n/p for n distinct values. This reference describes residue uniformity and is not a statistical null model or a p-value;

- `position_asymmetry_ncbi.csv` applies the Keto/Amino classification to each of the three positions of all 27 NCBI tables under Model 1 and Model 2. For T, P, N and Δ, it records directed Keto-minus-Amino differences, their residues modulo 37 and position-level summary flags;

- `representation_sensitivity.csv` recomputes 13 explicitly listed absolute and difference quantities under three representations of the encoded amino acid. Relative to the neutral free molecule, the proton and neutron offsets are $(0, 0)$ for the free molecule itself, $(-10, -8)$ for the peptide residue, and $(0, 0)$ for the zwitterion. For every quantity, the file records the result under each representation and its agreement with the free-molecule baseline.

These four files define the positional, modular, cross-table and representation-specificity controls. Their results are interpreted in Section 4.2.

Data and code availability. The accompanying repository contains the manuscript sources, five input datasets, 17 core computed datasets, four control datasets, the Monte Carlo summary table `mc_significance.csv`, four Python scripts and the interactive visualizer. Complete run commands and descriptions of the dataset columns are provided in the repository README: `https://github.com/RuslanKhafizov/genetic-code-arithmetic`. Versions of the materials are archived at Zenodo under the concept DOI 10.5281/zenodo.20360037.

3 Results

The arithmetic properties are considered here at three analytical levels. The first level (Section 3.1) comprises facts established without assigning values to the service codons.

The second level (Section 3.2) is the naive reading (Key 0), in which the three stop codons are assigned a contribution of zero and the initiator ATG is counted as methionine.

Section 3.3 reduces the asymmetry of the Keto and Amino branches to the nucleon differences of the Trp–Ile pair and examines the parity constraint on the proton quantities.

At the third level, Section 3.4 derives Key 1, the unique minimal non-negative integer assignment to the service codons determined by the joint conditions of nucleon-pool balance along the Keto/Amino axis and divisibility of the global pools by 37; equivalently, this balance augments the C/T symmetry imposed by the code with G/A symmetry in proton and neutron sums grouped by the nucleotide in the third position.

Section 3.5 uses this formal assignment to extend the arithmetic analysis to the complete 64-codon table and distinguishes between identities that follow directly from the derivation conditions and additional regularities not used to select Key 1.

Section 3.6 draws together the findings of all three levels.

The standard code remains the principal object of analysis throughout. Alternative NCBI translation tables are examined later for comparison and to assess the specificity of the results.

3.1 Parametrization-independent results

The parametrization-independent results for group-level nucleon quantities in this subsection are obtained under one of two conditions: either the summation domain is restricted to the service-excluded pool, or the group under consideration itself contains no service codons.

These results form the basis of the arithmetic structure that is traced further as the service codons are restored. The neutron sum of the service-excluded pool is divisible by 37, whereas the proton sum falls one unit short of the nearest multiple. Through the difference Δ this deviation is traced to the codons with G at the third position within Octet II — the group that contains the initiator ATG in the full table (Section 3.1.1). The nucleon sum of Octet I is divisible by 37, and all four of its quantities are divisible by 100 (Section 3.1.2). For the pyrimidine-restricted groups of the full code the difference Δ lies on the lattice of step 37, whereas the deviations arising on intersection with the octets cancel within the group (Section 3.1.3).

3.1.1 The service-excluded pool

The Keto/Amino and Strong/Weak partitions of $\mathcal{C}_{\mathrm{SE}}$ give pairwise equal neutron and proton sums, owing to pyrimidine synonymy (Section 2.4):

$$\begin{aligned} N_{\mathrm{SE}}(\mathrm{Keto}) &= N_{\mathrm{SE}}(\mathrm{Strong}) = 1813 = 49 \cdot 37, \\ N_{\mathrm{SE}}(\mathrm{Amino}) &= N_{\mathrm{SE}}(\mathrm{Weak}) = 1776 = 48 \cdot 37, \\ P_{\mathrm{SE}}(\mathrm{Keto}) &= P_{\mathrm{SE}}(\mathrm{Strong}) = 2108 = 57 \cdot 37 - 1, \\ P_{\mathrm{SE}}(\mathrm{Amino}) &= P_{\mathrm{SE}}(\mathrm{Weak}) = 2072 = 56 \cdot 37. \end{aligned}$$

Both neutron sums lie on the 37-lattice separately. The proton sum of the Amino half lands on it exactly, whereas the Keto half falls one unit short of the nearest multiple. Adding the halves gives, for the pool as a whole,

$$N(\mathcal{C}_{\mathrm{SE}}) = 3589 = 97 \cdot 37, \qquad P(\mathcal{C}_{\mathrm{SE}}) = 4180 = 113 \cdot 37 - 1, \qquad \Delta(\mathcal{C}_{\mathrm{SE}}) = 591 = 16 \cdot 37 - 1.$$

Analysis of Δ localizes the one-unit deviation of the proton sum of the pool from the nearest multiple of 37 in the group $\{\mathrm{G}\} \cap$ Octet II. On the Keto/Amino and Strong/Weak axes the neutron sums are multiples of 37, so $\Delta = P - N$ has the same residue as the proton sum. On the Purine/Pyrimidine axis neither P nor N lies on the lattice, but their residues coincide for Pyrimidine and differ by one for Purine:

$$\begin{aligned} \Delta_{\mathrm{SE}}(\mathrm{Keto}) = \Delta_{\mathrm{SE}}(\mathrm{Strong}) = \Delta_{\mathrm{SE}}(\mathrm{Purine}) &= 295 = 8 \cdot 37 - 1, \\ \Delta_{\mathrm{SE}}(\mathrm{Amino}) = \Delta_{\mathrm{SE}}(\mathrm{Weak}) = \Delta(\mathrm{Pyrimidine}) &= 296 = 8 \cdot 37. \end{aligned}$$

Keto, Strong, Amino, and Weak each contain 30 codons in the pool, Purine contains 28, and Pyrimidine contains 32. The same two values of Δ, 295 and 296, arise in all three partitions — both for parts of equal size 30/30 and for the unequal split 28/32.

At the level of the one-nucleotide groups none of the sums P and N is a multiple of 37, and the one-unit deviation in Δ is carried by one group — the branch $\{\mathrm{G}\}$, common to Keto, Strong, and Purine:

$$\Delta_{\mathrm{SE}}(\{\mathrm{G}\}) = 147 = 4 \cdot 37 - 1, \qquad \Delta_{\mathrm{SE}}(\{\mathrm{A}\}) = \Delta(\{\mathrm{C}\}) = \Delta(\{\mathrm{T}\}) = 148 = 4 \cdot 37.$$

Keto, Strong, and Purine are the unions of $\{\mathrm{G}\}$ with $\{\mathrm{T}\}$, $\{\mathrm{C}\}$, and $\{\mathrm{A}\}$ respectively. The one-unit deviation localized in $\Delta_{\mathrm{SE}}(\{\mathrm{G}\})$ can therefore be traced across all three axes.

Within Octet I all four one-nucleotide groups have $\Delta = 75$, so the difference falls entirely in Octet II:

$$\Delta_{\mathrm{SE}}(\{\mathrm{G}\} \cap \mathrm{Octet\ II}) = 72, \qquad \Delta_{\mathrm{SE}}(\{\mathrm{A}\} \cap \mathrm{Octet\ II}) = \Delta(\{\mathrm{C}\} \cap \mathrm{Octet\ II}) = \Delta(\{\mathrm{T}\} \cap \mathrm{Octet\ II}) = 73.$$

In the pool the groups $\{G\} \cap$ Octet II and $\{A\} \cap$ Octet II contain six codons each, whereas $\{C\} \cap$ Octet II and $\{T\} \cap$ Octet II contain eight. The observed distribution of the values 72 and 73 therefore cannot be attributed to group size.

On passing to the complete 64-codon table, these two six-codon branches are supplemented by different pairs of service codons: $\{A\} \cap$ Octet II by the stop codons TAA and TGA, and $\{G\} \cap$ Octet II by the stop codon TAG and the initiator ATG. As will be shown in the derivation of Key 1 (Section 3.4), the formal Δ-contribution under that assignment equals 1 for the initiator ATG and 0 for each stop codon. Therefore, under Key 1,

$$\Delta_1(\{G\} \cap \text{Octet II}) = 72 + 1 + 0 = 73, \qquad \Delta_1(\{A\} \cap \text{Octet II}) = 73 + 0 + 0 = 73.$$

The missing unit is thus returned through the initiator ATG, aligning the two branches. Since ATG belongs to the branch $\{G\}$ and to Keto, Strong, and Purine, the same single contribution simultaneously compensates for the deviation in all of these groups and in the value of Δ for the complete table under Key 1.

The full hierarchy of groups for the service-excluded pool, together with the corresponding nucleon quantities, is shown in Figure 2.

3.1.2 Arithmetic of Octet I

Octet I (id 12) contains no service codons, so its nucleon quantities are independent of any parametrization:

$$\begin{aligned} T(\text{Octet I}) &= 3700 = 100 \cdot 37, \quad P(\text{Octet I}) = 2000 = 2 \cdot 1000, \\ N(\text{Octet I}) &= 1700 = 17 \cdot 100, \quad \Delta(\text{Octet I}) = 300 = 3 \cdot 100. \end{aligned}$$

All four quantities are multiples of 100, while the proton sum $P(\text{Octet I})$ is also a multiple of 1000. In addition, the nucleon sum $T(\text{Octet I})$ is divisible by 37.

Octet I consists of eight fully four-fold degenerate codon families. Each of its six 16-codon groups (ids 13–18) contains two codons from every family, so each of its nucleon quantities equals half the corresponding quantity of the whole octet:

$$T = 1850 = 50 \cdot 37, \qquad P = 1000 = 10^3, \qquad N = 850, \qquad \Delta = 150.$$

Each of its four 8-codon groups (ids 19–22) contains one codon from every family, so its quantities equal one quarter of the whole-octet values:

$$T = 925 = 25 \cdot 37, \qquad P = 500 = 5 \cdot 100, \qquad N = 425, \qquad \Delta = 75 = 2 \cdot 37 + 1.$$

Thus, at all three levels of the Octet I hierarchy, exact scaling preserves divisibility of the nucleon sum T by 37 and a single rational profile with a common denominator of 37:

$$\frac{P}{T} = \frac{20}{37}, \qquad \frac{N}{T} = \frac{17}{37}, \qquad \frac{\Delta}{T} = \frac{3}{37}.$$

The proton and neutron sums of the whole octet have residues $+2$ and -2 modulo 37, which cancel in $T = P + N$. The neutron residue -2 of Octet I is, in turn, compensated by the residue $+2$ of the Octet II neutron sum in the service-excluded pool. Adding the neutron sums of the two octets gives the pool neutron sum $N(\mathcal{C}_{\text{SE}})$ obtained in Section 3.1.1. Both cancellations are represented by the following equalities:

$$\begin{aligned} P(\text{Octet I}) &= 2000 = 54 \cdot 37 + 2, \\ N(\text{Octet I}) &= 1700 = 46 \cdot 37 - 2, \\ N_{\text{SE}}(\text{Octet II}) &= 1889 = 51 \cdot 37 + 2, \\ N(\mathcal{C}_{\text{SE}}) &= N(\text{Octet I}) + N_{\text{SE}}(\text{Octet II}) \\ &= 1700 + 1889 = 3589 = 97 \cdot 37. \end{aligned}$$

Figure 2: Structure of the service-codon-excluded 60-codon pool $\mathcal{C}_{\mathrm{SE}}$ (`pool60`). For each group, all four quantities are computed using only the non-service codons it contains; the three stop codons and the initiator ATG are excluded, so every quantity shown is independent of parametrization. The quantities shown are the total nucleon sum T, proton sum P, neutron sum N, and hydrogen atom count $\Delta = P - N$. Cell layout: the group name appears in the upper-left corner, the number of codons remaining in the group in the upper-right corner, and the identifier from `codon_groups.csv` in the lower-right corner; twin pairs of groups with identical values are collapsed into a single cell, with the second name in the lower-left corner and both identifiers listed, separated by a comma, in the lower-right corner. Green, blue, and purple highlights indicate divisibility by 37, $111 = 3 \cdot 37$, and $999 = 27 \cdot 37$, respectively; the largest applicable modulus is marked. Figures 3 and 4 use the same grid and color code, but represent the complete 64-codon table. All three figures were generated by the `visualization.html` script.

At the 16-codon level, exact halving gives residues $+1$ for P and -1 for N modulo 37. Of the four 8-codon groups, the two pyrimidine groups— $\{C\} \cap$ Octet I (id 19) and $\{T\} \cap$ Octet I (id 21)—form the Pyrimidine group within Octet I (id 18), which enters the pyrimidine cascade in Section 3.1.3.

3.1.3 Arithmetic of pyrimidine-restricted groups

By pyrimidine synonymy (Section 2.4), the $\{C\}$ and $\{T\}$ groups have identical nucleon quantities at the full-code level and after intersection with either octet. Together they form the Pyrimidine group (id 7). These groups and their octet intersections contain no service codons, so their values are independent of parametrization. Unlike the Octet I hierarchy, the full Pyrimidine group spans both octets and includes partially degenerate codon families. Therefore, not all values at its lower levels are obtained by a single exact scaling; they are examined separately below.

For the full Pyrimidine group, the nucleon quantities are

$$\begin{aligned} T(\text{Pyrimidine}) &= 4096 = 2^{12}, & P(\text{Pyrimidine}) &= 2196, \\ N(\text{Pyrimidine}) &= 1900 = 19 \cdot 100, & \Delta(\text{Pyrimidine}) &= 296 = 8 \cdot 37. \end{aligned}$$

Here, Δ satisfies the primary preselected criterion of divisibility by 37, and $N = 1900$ satisfies the roundness criterion. The exact power of two $T = 2^{12}$ is an additional descriptive observation.

By the C/T identity, the two 16-codon groups $\{C\}$ (id 8) and $\{T\}$ (id 10) have identical nucleon quantities. Because they partition the Pyrimidine group, each value is half the corresponding full-group value:

$$\begin{aligned} T(\{C\}) = T(\{T\}) &= 2048 = 2^{11}, & P(\{C\}) = P(\{T\}) &= 1098, \\ N(\{C\}) = N(\{T\}) &= 950, & \Delta(\{C\}) = \Delta(\{T\}) &= 148 = 4 \cdot 37. \end{aligned}$$

The entire profile at this level is therefore obtained by exact scaling and does not form a separate set of independent regularities.

At the 8-codon level, all four Octet I groups, including $\{C\} \cap$ Octet I (id 19) and $\{T\} \cap$ Octet I (id 21), have $\Delta = 75$ (Section 3.1.2). The corresponding $\{C\} \cap$ Octet II (id 30) and $\{T\} \cap$ Octet II (id 32) groups have $\Delta = 73$. Their unions within the octets form the Pyrimidine $\cap$ Octet I (id 18) and Pyrimidine $\cap$ Octet II (id 29) groups. These values form the following cascade:

$$\begin{aligned} \Delta(\{C\} \cap \text{Octet I}) = \Delta(\{T\} \cap \text{Octet I}) &= 75 = 2 \cdot 37 + 1, \\ \Delta(\{C\} \cap \text{Octet II}) = \Delta(\{T\} \cap \text{Octet II}) &= 73 = 2 \cdot 37 - 1, \\ \Delta(\text{Pyrimidine} \cap \text{Octet I}) &= 2 \cdot 75 = 150 = 4 \cdot 37 + 2, \\ \Delta(\text{Pyrimidine} \cap \text{Octet II}) &= 2 \cdot 73 = 146 = 4 \cdot 37 - 2, \\ \Delta(\text{Pyrimidine}) &= 150 + 146 = 296 = 8 \cdot 37. \end{aligned}$$

The one-unit deviations of opposite sign, $+1$ and -1, double to $+2$ and -2 when the groups within each octet are added, and cancel when the octets are added. Thus, the values 150, 146, and 296 belong to one additive scheme rather than constituting independent results.

The relation of the pyrimidine cascade to the pattern of one-unit deviations in the proton and neutron sums is considered under Key 0 (Section 3.2.3).

The pyrimidine cascade should also be distinguished from the one-unit deviation of $\Delta_{\text{SE}}(\{G\})$ from the lattice of step 37 (Section 3.1.1). The pyrimidine deviations cancel between the octets, whereas the deviation of the $\{G\}$ group is not compensated when it is combined with another one-nucleotide group and propagates to the Keto, Strong, and Purine groups. Because the neutron sums of Keto and Strong in the service-excluded pool are multiples of 37, the identity $P = N + \Delta$ implies that their proton sums have the same one-unit deviation, -1.

3.2 Results under Key 0

Under the naive reading (Key 0), summation is performed over the complete 64-codon table: the stop-codon contribution is zero, whereas the initiator ATG is counted as methionine (Section 2.5). Thus, in passing from the service-excluded pool to the complete table, the only nonzero contribution comes from ATG:

$$\big(T(\text{Met}), P(\text{Met}), N(\text{Met}), \Delta(\text{Met})\big) = (149, 80, 69, 11).$$

The deviations from the nearest multiples of 37 for the service-excluded pool, methionine, and the complete 64-codon table are related as

$$(-1, -1, 0, -1) + (+1, +6, -5, +11) = (0, +5, -5, +10).$$

The ATG contribution brings the nucleon sum T exactly onto the 37-lattice, since $149 = 4 \cdot 37 + 1$ exactly compensates the one-unit deviation -1, but at the same time moves the neutron sum off the lattice, although it is divisible by 37 in the pool (Section 3.1.1).

Below, we show how this transformation propagates through the nested groups containing ATG. In the Amino and Weak groups, all the restored positions are occupied by stop codons, so the sums of the complete 32-codon groups coincide numerically with the sums over the 30 codons in the corresponding branches of the service-excluded pool. These sums already lie on the 37-lattice (Section 3.1.1), and the divisibility is therefore preserved in passing to the complete table. Under Key 0, restoring the service positions makes the groups on the two sides of the mirror pattern of one-unit deviations equal in size. The pattern appears at two levels: at the 16-codon level in the proton and neutron sums, and at the 8-codon level in the difference Δ. Within Octet II at the 8-codon level, this pattern, established earlier for the pyrimidine groups (Section 3.1.3), also extends to the {A} group: its value of Δ already coincided with theirs in the pool, while restoring the service positions makes all three groups equal in size. The sole exception is the {G} branch, which contains ATG.

The overall pattern of group quantities under Key 0 is shown in Fig. 3.

3.2.1 Divisibility of T by 37 at the upper levels of partitioning

The ATG contribution falls in Octet II. Under Key 0, its four 8-codon groups have the following profile:

$$\begin{aligned} T(\{\text{C}\} \cap \text{Octet II}) = T(\{\text{T}\} \cap \text{Octet II}) &= 1123 = 30 \cdot 37 + 13, \\ T_0(\{\text{A}\} \cap \text{Octet II}) &= 875 = 24 \cdot 37 - 13, \\ T_0(\{\text{G}\} \cap \text{Octet II}) &= 1097 = 30 \cdot 37 - 13. \end{aligned}$$

The pyrimidine branches deviate from the nearest multiple by $+13$, the purine branches by -13. In the service-excluded pool, the deviation of the {G} branch differs by one unit from that of the {A} branch:

$$T_{\text{SE}}(\{\text{G}\} \cap \text{Octet II}) = 948 = 26 \cdot 37 - 14.$$

The ATG contribution changes the deviation of the {G} branch from -14 to -13, so that both purine branches have the same deviation.

Among the groups of Octet II, the nucleon sum is divisible by 37 exactly for those that contain equal numbers of purine and pyrimidine branches:

$$\begin{aligned} T_0(\text{Keto} \cap \text{Octet II}) = T_0(\text{Strong} \cap \text{Octet II}) &= 2220 = 60 \cdot 37, \\ T_0(\text{Amino} \cap \text{Octet II}) = T_0(\text{Weak} \cap \text{Octet II}) &= 1998 = 54 \cdot 37, \\ T_0(\text{Octet II}) &= 4218 = 114 \cdot 37. \end{aligned}$$

The intersections of the Purine and Pyrimidine groups with Octet II consist of branches of a single type, so the branch deviations do not cancel but double relative to the sums of the corresponding multiples of 37.

Figure 3: Nucleon quantities T, P, N, and Δ for all codon groups under Key 0. The grid, twin-group pair merging, and divisibility color code are the same as in Fig. 2, but the quantities are calculated for the complete 64-codon table. Generated with `visualization.html`.

Each 16-codon half of a chemical axis in Octet I has $T = 1850 = 50 \cdot 37$ (Section 3.1.2). Adding the corresponding half carries the divisibility over to the complete groups of the Keto/Amino and Strong/Weak axes, and adding the two octets carries it over to the whole table:

$$\begin{aligned} T_0(\text{Keto}) = T_0(\text{Strong}) &= 1850 + 2220 = 4070 = 110 \cdot 37, \\ T_0(\text{Amino}) = T_0(\text{Weak}) &= 1850 + 1998 = 3848 = 104 \cdot 37, \\ T_0(\text{All}) &= 3700 + 4218 = 7918 = 214 \cdot 37. \end{aligned}$$

The divisibilities listed follow from the mirror deviations $+13/-13$ between the branches of Octet II and from the divisibility of Octet I, and do not form an independent set of results.

The divisibilities $T(\text{Octet I}) = 3700$, $T_0(\text{Octet II}) = 4218$, and $T_0(\text{Keto} \cap \text{Octet II}) = T_0(\text{Strong} \cap \text{Octet II}) = 2220$ correspond, in the group formalism adopted here, to results of Shcherbak and Makukov [8] and Panov and Filatov [5]; see Section 4.4 for further discussion. The complete values of the group quantities under Key 0 are given in `key0_nucleon-data.csv`.

3.2.2 Alignment of Amino and Weak with the lattice of step 37

Among the six 32-codon groups of the three chemical axes, only Amino (id 4) and Weak (id 5) have all four nucleon quantities divisible by 37:

$$\begin{aligned} T_0(\text{Amino}) = T_0(\text{Weak}) &= 3848 = 104 \cdot 37, \\ P_0(\text{Amino}) = P_0(\text{Weak}) &= 2072 = 56 \cdot 37, \\ N_0(\text{Amino}) = N_0(\text{Weak}) &= 1776 = 48 \cdot 37, \\ \Delta_0(\text{Amino}) = \Delta_0(\text{Weak}) &= 296 = 8 \cdot 37. \end{aligned}$$

The divisibility of T was obtained above. In Octet I the halves of both groups have residues $+1$ for P and -1 for N (Section 3.1.2), and in Octet II the opposite:

$$\begin{aligned} P(\text{Amino} \cap \text{Octet I}) = P(\text{Weak} \cap \text{Octet I}) &= 1000 = 27 \cdot 37 + 1, \\ P_0(\text{Amino} \cap \text{Octet II}) = P_0(\text{Weak} \cap \text{Octet II}) &= 1072 = 29 \cdot 37 - 1, \\ N(\text{Amino} \cap \text{Octet I}) = N(\text{Weak} \cap \text{Octet I}) &= 850 = 23 \cdot 37 - 1, \\ N_0(\text{Amino} \cap \text{Octet II}) = N_0(\text{Weak} \cap \text{Octet II}) &= 926 = 25 \cdot 37 + 1. \end{aligned}$$

When the halves are added, these residues cancel each other, and with them the residue of the difference $\Delta = P - N$.

For the Amino and Weak groups, the ratios to the nucleon sum form a profile with a common denominator of 13:

$$\frac{P}{T} = \frac{7}{13}, \qquad \frac{N}{T} = \frac{6}{13}, \qquad \frac{\Delta}{T} = \frac{1}{13}.$$

3.2.3 One-unit deviations in Δ and the {G} branch

In Octet I, all four branches have $\Delta = 75 = 2 \cdot 37 + 1$ (Section 3.1.2). In Octet II, the deviation has the opposite sign and, under Key 0, occurs in three of the four branches:

$$\begin{aligned} \Delta(\{\text{C}\} \cap \text{Octet II}) = \Delta(\{\text{T}\} \cap \text{Octet II}) = \Delta_0(\{\text{A}\} \cap \text{Octet II}) &= 73 = 2 \cdot 37 - 1, \\ \Delta_0(\{\text{G}\} \cap \text{Octet II}) &= 83 = 2 \cdot 37 + 9. \end{aligned}$$

The equality of the {C} and {T} branches follows from pyrimidine synonymy (Section 3.1.3), whereas their equality with the purine branch {A} is an additional numerical fact. Under Key 0, its two restored positions are occupied by stop codons, so the pool value $\Delta = 73$ is preserved while the group size increases to eight codons. The {G} branch already differs in the service-excluded pool, where it has $\Delta = 72$ rather than 73 in the other branches; the ATG contribution changes this value to 83 (Section 3.1.1).

Adding the Octet I branches gives the values of the one-nucleotide groups in the complete table:

$$\Delta(\{\mathrm{C}\}) = \Delta(\{\mathrm{T}\}) = \Delta_0(\{\mathrm{A}\}) = 148 = 4 \cdot 37, \qquad \Delta_0(\{\mathrm{G}\}) = 158 = 4 \cdot 37 + 10.$$

The {G} branch belongs to Keto, Strong, and Purine, whereas the other three one-nucleotide groups lie on the lattice. The residue +10 therefore propagates to all three axes and to the complete table:

$$\Delta_0(\mathrm{Keto}) = \Delta_0(\mathrm{Strong}) = \Delta_0(\mathrm{Purine}) = 306 = 8 \cdot 37 + 10,$$
$$\Delta_0(\mathrm{All}) = 602 = 16 \cdot 37 + 10.$$

The deviation of the complete table in Δ, reported at the beginning of this section, is thus localized in the {G} branch.

The residues of P and N at the 16-codon level also follow from the 8-codon pattern (Section 3.2.2). In the halves of the Amino and Weak groups, two residues $\Delta \equiv +1$ in Octet I add to $+2$, whereas two residues $\Delta \equiv -1$ in Octet II add to -2. Because the corresponding sums T are divisible by 37, $2P = T + \Delta$ and $2N = T - \Delta$ give $(P, N) \equiv (+1, -1) \pmod{37}$ in Octet I and $(P, N) \equiv (-1, +1) \pmod{37}$ in Octet II.

3.3 Localization of the Keto/Amino asymmetry and proton parity

The asymmetry between the Keto and Amino branches is visible at both levels considered here. It is shown first for the full table under Key 0, where it reduces to the nucleon quantities of three molecules, and then for the service-excluded pool, where two remain.

3.3.1 Localization of the Keto/Amino asymmetry

For the full table under Key 0, the differences between the nucleon quantities of the Keto and Amino branches are

$$\begin{aligned}
T_0(\mathrm{Keto}) - T_0(\mathrm{Amino}) &= 4070 - 3848 = 222 = 6 \cdot 37,\\
P_0(\mathrm{Keto}) - P_0(\mathrm{Amino}) &= 2188 - 2072 = 116,\\
N_0(\mathrm{Keto}) - N_0(\mathrm{Amino}) &= 1882 - 1776 = 106,\\
\Delta_0(\mathrm{Keto}) - \Delta_0(\mathrm{Amino}) &= 306 - 296 = 10.
\end{aligned}$$

Reduction at the family level. By pyrimidine synonymy (Section 2.4), in all 16 codon families the codons NNT and NNC encode the same amino acid. In the 13 families that contain no service codons, the products of NNG and NNA also coincide. Therefore, the Keto branch, represented by the pair NNG and NNT, and the Amino branch, represented by the pair NNA and NNC, make equal nucleon contributions regardless of which amino acids the family encodes.

The TA family also makes equal contributions: its pyrimidine pair TAC and TAT consists of two Tyr codons, whereas its purine pair TAG and TAA consists of two stop codons, one in each branch. The latter cancel when these two stop codons receive equal assignments.

The asymmetry remains in two families, and in each of them one codon pair cancels whereas the other does not. In AT, ATT and ATC cancel, encoding Ile in the Keto and Amino branches; the remaining pair is ATG/ATA: the initiator versus the third Ile codon. In TG, TGT and TGC cancel, encoding Cys; the remaining pair is TGG/TGA: a Trp codon versus a stop codon.

Under Key 0, the initiator is counted as the methionine it encodes, whereas the stop codon TGA contributes zero. Therefore, from the two remaining pairs, for $X \in \{T, P, N, \Delta\}$,

$$X_0(\mathrm{Keto}) - X_0(\mathrm{Amino}) = X(\mathrm{Trp}) + X(\mathrm{Met}) - X(\mathrm{Ile}),$$

so the asymmetry of the full table reduces to the nucleon quantities of three molecules; in particular,

$$T_0(\mathrm{Keto}) - T_0(\mathrm{Amino}) = 204 + 149 - 131 = 222 = 6 \cdot 37.$$

Reduction to the Trp–Ile **pair.** The initiator ATG is the only service codon that contributes to these differences under Key 0. In the service-excluded pool, it is removed from the sums together with the three stop codons, and the asymmetry reduces to the nucleon differences of a single molecular pair, Trp–Ile:

$$\begin{aligned} T_{\mathrm{SE}}(\mathrm{Keto}) - T_{\mathrm{SE}}(\mathrm{Amino}) &= 3921 - 3848 = 73, \\ P_{\mathrm{SE}}(\mathrm{Keto}) - P_{\mathrm{SE}}(\mathrm{Amino}) &= 2108 - 2072 = 36, \\ N_{\mathrm{SE}}(\mathrm{Keto}) - N_{\mathrm{SE}}(\mathrm{Amino}) &= 1813 - 1776 = 37, \\ \Delta_{\mathrm{SE}}(\mathrm{Keto}) - \Delta_{\mathrm{SE}}(\mathrm{Amino}) &= 295 - 296 = -1. \end{aligned}$$

Thus, for $X \in \{T, P, N, \Delta\}$,

$$X_{\mathrm{SE}}(\mathrm{Keto}) - X_{\mathrm{SE}}(\mathrm{Amino}) = X(\mathrm{Trp}) - X(\mathrm{Ile}).$$

This reduction is determined by the synonymy structure and the positions of the service codons.

By the twin-group identities (Section 2.4), $Q_{\mathrm{SE}}(\mathrm{Strong}) = Q_{\mathrm{SE}}(\mathrm{Keto})$ and $Q_{\mathrm{SE}}(\mathrm{Weak}) = Q_{\mathrm{SE}}(\mathrm{Amino})$ for $Q \in \{T, P, N, \Delta\}$. Therefore, the Strong/Weak-axis asymmetry is described by the same differences as the Keto/Amino asymmetry and likewise reduces to the Trp–Ile pair.

Local and global scales. For the pair localized above, the corresponding differences follow directly from the nucleon counts of the amino acids:

$$N(\mathrm{Trp}) - N(\mathrm{Ile}) = 96 - 59 = 37, \qquad P(\mathrm{Trp}) - P(\mathrm{Ile}) = 108 - 72 = 36.$$

The proton and neutron differences differ by one unit, corresponding to one hydrogen atom: tryptophan contains 12 hydrogen atoms, whereas isoleucine contains 13.

The same residues modulo 37 were previously established at the global level for the entire service-excluded pool (Section 3.1.1):

	global	local
N	0	0
P	-1	-1

The global and local facts do not follow from one another: the former refer to the entire service-excluded pool, whereas the latter refer to the nucleon profile of the Trp–Ile pair to which the Keto/Amino asymmetry reduces in this pool.

In the following analysis, the local differences enter the balance conditions for the service codons, whereas the global residues enter the divisibility conditions in the analytical derivation of Key 1 (Section 3.4).

Arithmetic rarity and structural uniqueness of the Trp–Ile pair. The neutron and proton differences of this pair are rare within the fixed set of canonical amino acids.

The value $|\delta N| = 37$*.* Among the 190 pairs of canonical amino acids, the value $|\delta N| = 37$ occurs for only two pairs: Trp–Ile and Trp–Leu. The latter gives the same value because Leu and Ile are isomers with identical nucleon counts, $P = 72$ and $N = 59$; the pair structurally realized through the AT and TG families is Trp–Ile. The complete distribution of $|\delta N|$ over the 190 pairs is provided in `amino_acids_neutron_differences.csv`.

The value $|\delta P| = 36$*.* Among the 190 pairs of canonical amino acids, the value $|\delta P| = 36$ occurs only for the same two pairs, Trp–Ile and Trp–Leu. The complete distribution of $|\delta P|$ over the 190 pairs is provided in `amino_acids_proton_differences.csv`.

The value $|\delta T| = 73$*.* For the Trp–Ile pair, this value follows from the proton and neutron differences given above, which have the same sign. Among all 190 pairs of canonical amino acids, however, $|\delta T| = 73$ occurs only for Trp–Ile and Trp–Leu. The complete distribution of $|\delta T|$ is provided in `amino_acids_nucleon_differences.csv`.

3.3.2 Proton parity

Each of the 20 canonical amino acids has an even total proton count: all P values in `amino_acids_nucleons.csv` are even. In their electronic ground states, the neutral molecules of the canonical amino acids are closed-shell systems in which electrons occupy orbitals in pairs; therefore, their total number of electrons is even. Because in a neutral molecule the number of electrons equals the total number of protons in its nuclei, the parity of P follows.

Consequently, proton sums over amino acids and pairwise proton differences are even, and none of these quantities can equal an odd multiple of 37. Thus, $|\delta P(\mathrm{Trp}, \mathrm{Ile})|$ cannot equal 37: the nearest even values to 37 are 36 and 38, and the actual difference for the localized pair is 36. For the same reason, the proton sum of the service-excluded pool cannot equal the next higher multiple $113 \cdot 37$, which is odd (Section 3.1.1).

Parity imposes only a constraint. It determines neither the direction nor the magnitude of the deviation: the fact that both the local difference and the global sum lie one unit below the corresponding multiple of 37 follows from the specific nucleon counts.

3.4 Analytical derivation of Key 1

The results of Sections 3.1 and 3.2 show that whether the arithmetic regularities appear depends on which service codons a group contains.

Octet I and the pyrimidine-restricted groups contain no service codons at all, and they exhibit arithmetic order (Sections 3.1.2 and 3.1.3). The Amino and Weak branches contain stop codons but not the initiator ATG: all four of their nucleon quantities are divisible by 37 already in the service-excluded pool, and the zero contribution of the stop codons leaves them unchanged, so the divisibility is preserved under Key 0 (Section 3.2.2). In the Keto and Strong branches, which contain the initiator ATG, the proton and neutron sums and the difference Δ deviate under Key 0 from the 37-lattice. The deviation in Δ is localized in the $\{G\}$ branch, which contains ATG: the other three one-nucleotide groups share a single value of Δ, divisible by 37 (Section 3.2.3).

This points to the possibility of a different parametrization of the service codons, one that reconciles the arithmetic regularities of the groups considered. In this section we derive such a parametrization analytically from conditions of symmetry, divisibility, and minimality. The resulting assignment serves as a formal device for reconciling the sums of the complete table; the underlying arithmetic regularities are established in the service-excluded pool. The specificity of the result across natural codes is examined in Section 4.1.

For the assignment K sought, let $(p_{\mathrm{stop}}, n_{\mathrm{stop}})$ denote the common parameters of the three stop codons and $(p_{\mathrm{start}}, n_{\mathrm{start}})$ those of the initiator ATG. All parameters are non-negative integers. The total contribution of the service codons is minimized separately for neutrons and protons, as defined in Section 2.5. We first determine the neutron parameters, then the proton parameters.

3.4.1 Determination of the neutron parameters

Distribution of the service codons. All four service codons belong to Octet II, so the entire added contribution falls in that octet. The initiator ATG and the stop codon TAG have G in the third position and belong to the Keto branch; the stop codons TAA and TGA have A in the third position and belong to the Amino branch. Thus Keto contains one stop codon and the initiator, whereas Amino contains two stop codons. The same distribution of service positions holds for Strong and Weak. By the twin-group identities (Section 2.4), Keto/Amino balance also ensures Strong/Weak balance.

Balance equation for Keto and Amino. The complete neutron sums consist of the service-excluded pool sums (Section 3.3.1) and the unknown contributions:

$$\begin{aligned} N_K(\text{Keto}) &= 1813 + n_{\text{start}} + n_{\text{stop}}, \\ N_K(\text{Amino}) &= 1776 + 2\,n_{\text{stop}}. \end{aligned}$$

The condition $N_K(\text{Keto}) = N_K(\text{Amino})$ gives

$$n_{\text{stop}} - n_{\text{start}} = 37.$$

The right-hand side is the difference between the neutron counts of tryptophan and isoleucine, to which the pool imbalance has been reduced. Balance fixes the difference between the parameters, leaving a one-parameter family. In the main derivation, the next condition is divisibility of the total sum. Its compatibility with the choice of chemical axis is examined in Section 3.4.3.

Divisibility of the total neutron sum by 37. The service-excluded pool has $N(\mathcal{C}_{\text{SE}}) = 3589 = 97 \cdot 37$ (Section 3.1.1). Under Key 0, the initiator contributes the neutron count of methionine, giving

$$N_0(\text{All}) = 3589 + 69 = 3658 = 99 \cdot 37 - 5.$$

To preserve divisibility in the complete table, we require

$$N_K(\text{All}) = 3589 + n_{\text{start}} + 3\,n_{\text{stop}} \equiv 0 \pmod{37},$$

or equivalently

$$n_{\text{start}} + 3\,n_{\text{stop}} \equiv 0 \pmod{37}.$$

Solution. Substituting $n_{\text{stop}} = n_{\text{start}} + 37$ from the first condition into the second yields

$$4\,n_{\text{start}} + 111 \equiv 0 \pmod{37}.$$

Since $111 = 3 \cdot 37 \equiv 0 \pmod{37}$ and $\gcd(4, 37) = 1$, this reduces to

$$n_{\text{start}} \equiv 0 \pmod{37},$$

giving the family of non-negative integer solutions $n_{\text{start}} = 0, 37, 74, \ldots$. The minimal solution is

$$n_{\text{start}} = 0, \quad n_{\text{stop}} = 37.$$

The divisibility requirement does not change the minimum just obtained. With non-negative integer parameters, the Keto/Amino balance alone admits $n_{\text{start}} = 0, 1, 2, \ldots$ with $n_{\text{stop}} = n_{\text{start}} + 37$; divisibility narrows this family to $n_{\text{start}} = 0, 37, 74, \ldots$, but in both cases the minimal total added contribution is attained at $n_{\text{start}} = 0$ and $n_{\text{stop}} = 37$. For the proton component the divisibility requirement does change the minimum. The structure of the solution families and the role of each condition are addressed in Section 3.4.3.

Inputs and the result of the derivation. The system uses the local difference 37 and the total neutron sum of the pool, $3589 = 97 \cdot 37$. Both quantities are established before assigning values to the service codons. The coefficient 4 in the combined congruence arises when the balance equation is substituted into the sum of the contributions of three stop codons and the initiator. Coprimality of 4 and 37 allows the congruence to be reduced; this is a property of the equation, and the parameters obtained constitute its minimal solution.

Coincidence with the Octet I sum. After determining the parameters, the total neutron sum is

$$N_1(\text{All}) = 3589 + 0 + 3 \cdot 37 = 3700 = T(\text{Octet I}).$$

Equality with the nucleon sum of Octet I was not used as a condition of the derivation. It already holds at the minimal neutron assignment satisfying balance alone. At the level of Octet II, the same relation takes the form

$$N_1(\text{Octet II}) = 1889 + 0 + 3 \cdot 37 = 2000 = P(\text{Octet I}).$$

This restates the preceding equality after subtracting $N(\text{Octet I}) = 1700$, rather than constituting a separate coincidence. Balance and the twin-group identities give

$$\begin{aligned} N_1(\text{Keto}) = N_1(\text{Amino}) = N_1(\text{Strong}) &= N_1(\text{Weak}) \\ &= 1850 = \tfrac{1}{2}\,T(\text{Octet I}). \end{aligned}$$

Here 1850 is obtained by halving the total sum just found. Further consequences are considered in Section 3.5.

3.4.2 Determination of the proton parameters

The proton parameters are determined separately from the neutron parameters under the same conditions: Keto/Amino balance, divisibility of the total sum by 37, and minimality. Equality of the proton and neutron stop parameters is not assumed.

Balance equation for Keto and Amino. From the service-excluded pool sums (Section 3.3.1), we obtain

$$\begin{aligned} P_K(\text{Keto}) &= 2108 + p_{\text{start}} + p_{\text{stop}}, \\ P_K(\text{Amino}) &= 2072 + 2\,p_{\text{stop}}. \end{aligned}$$

The condition $P_K(\text{Keto}) = P_K(\text{Amino})$ gives

$$p_{\text{stop}} - p_{\text{start}} = 36.$$

The right-hand side equals $P(\text{Trp}) - P(\text{Ile})$. Parity rules out 37 but does not determine the actual difference 36, which follows from the proton counts of these molecules (Section 3.3.2).

Divisibility of the total proton sum by 37. The proton sum of the service-excluded pool is

$$\begin{aligned} P(\mathcal{C}_{\text{SE}}) &= P(\text{Octet I}) + P_{\text{SE}}(\text{Octet II}) \\ &= 2000 + 2180 = 4180 = 113 \cdot 37 - 1. \end{aligned}$$

This is a one-unit deviation from the nearest multiple of 37. For the complete table, we require

$$P_K(\text{All}) = 4180 + p_{\text{start}} + 3\,p_{\text{stop}} \equiv 0 \pmod{37},$$

or equivalently

$$p_{\text{start}} + 3\,p_{\text{stop}} \equiv 1 \pmod{37}.$$

Solution. Substituting $p_{\text{stop}} = p_{\text{start}} + 36$ gives

$$4\,p_{\text{start}} + 108 \equiv 1 \pmod{37}.$$

Since $108 = 3 \cdot 37 - 3$ and $\gcd(4, 37) = 1$,

$$4\,p_{\text{start}} \equiv 4 \pmod{37}, \qquad p_{\text{start}} \equiv 1 \pmod{37}.$$

The non-negative integer solutions have $p_{\text{start}} = 1, 38, 75, \ldots$. The minimal solution is

$$p_{\text{start}} = 1, \qquad p_{\text{stop}} = 37.$$

The family structure and the minimality of this choice are considered in Section 3.4.3.

The resulting equality of the stop parameters. The proton system uses the local difference 36 and the one-unit deviation of the total pool sum from a multiple of 37. As in the neutron derivation, the coefficient 4 is determined by the distribution of service positions and the balance equation. Reducing the congruence and selecting the minimum yields parameters that can be compared with the neutron solution (Section 3.4.1):

$$p_{\text{stop}} = n_{\text{stop}} = 37.$$

Equality of the proton and neutron parameters was not among the conditions of the two systems. It arises when their minimal non-negative integer solutions are selected.

Macroscopic consequences. Under Key 1, the total proton sum is 4292, while balance and the twin-group identities give

$$\begin{aligned} P_1(\text{Keto}) = P_1(\text{Amino}) = P_1(\text{Strong}) &= P_1(\text{Weak}) \\ &= 2146 = \tfrac{1}{2}\,T_1(\text{Octet II}). \end{aligned}$$

The corresponding complete-table equality is

$$P_1(\text{All}) = T_1(\text{Octet II}) = 4292.$$

The nucleon sum of Octet II here is computed under Key 1 and depends on the service-codon assignment. This equality follows from $N_1(\text{Octet II}) = P(\text{Octet I})$ and the definition $T = P + N$: it expresses the same relation between the complete table and the octets as the neutron coincidence. The full analysis is given in Section 3.5.

3.4.3 Hierarchy of constraints and the choice of Key 1

The systems in Sections 3.4.1 and 3.4.2 share the same construction. For non-negative integer parameters, the assignment is selected in three steps:

1. Keto/Amino balance fixes the differences $n_{\text{stop}} - n_{\text{start}} = 37$ and $p_{\text{stop}} - p_{\text{start}} = 36$. Two free parameters remain: the neutron and proton values assigned to the initiator. Every member of this family equalizes the four 32-codon groups Keto, Amino, Strong, and Weak in P and N, and hence in T and Δ.

2. Divisibility of the total sums by 37 restricts the initiator values to $n_{\text{start}} = 37k$ and $p_{\text{start}} = 1 + 37m$, where k, m are non-negative integers. The family still has two free integer parameters, but the step in each direction increases to 37.

3. The total added contributions are $111 + 148k$ for neutrons and $112 + 148m$ for protons. They increase strictly with k and m, respectively, so the unique minimum is attained at $k = m = 0$, giving Key 1: $(p_{\text{stop}}, n_{\text{stop}}) = (37, 37)$, $(p_{\text{start}}, n_{\text{start}}) = (1, 0)$.

Uniqueness refers to the minimum under the stated conditions: balance and divisibility also admit larger assignments. In particular, equality of the proton and neutron stop parameters holds whenever $k = m$, but their common value equals 37 only at $k = m = 0$.

The contribution of the divisibility condition. We compare the assignments with the minimal total added contribution, obtained separately for protons and for neutrons, under the Keto/Amino balance alone and under the additional requirement of divisibility by 37; in both cases the parameters are non-negative integers:

$$\begin{aligned} \text{balance only:} \quad & (p_{\text{stop}}, n_{\text{stop}}) = (36, 37), \quad (p_{\text{start}}, n_{\text{start}}) = (0, 0), \\ \text{balance and divisibility:} \quad & (p_{\text{stop}}, n_{\text{stop}}) = (37, 37), \quad (p_{\text{start}}, n_{\text{start}}) = (1, 0). \end{aligned}$$

The divisibility requirement does not change the minimal neutron assignment: the neutron sum of the complete table equals 3700 and is divisible by 37 under the balance alone. Under the balance alone the minimal proton sum is $4288 = 116 \cdot 37 - 4$; adding one proton to each of the four service positions raises it to $4292 = 116 \cdot 37$. The balance is preserved, since each half of the Keto/Amino axis contains two service positions. The additional divisibility requirement therefore increases the minimal contribution by four protons and requires no additional neutrons.

The same increment removes the difference in Δ across the Purine/Pyrimidine axis. For the Pyrimidine branch $\Delta = 296$ is independent of the parametrization (Section 3.1.3), whereas under the minimal assignment that secures the Keto/Amino balance alone the Purine branch has $\Delta = 292$: each stop codon carries $\Delta = 36 - 37 = -1$, which shifts the service-excluded pool value 295 (Section 3.1.1) down by three units. All four service codons belong to the Purine branch, so adding one proton to each of them equalizes the values of the two branches. An alternative derivation is considered below, in which this equality is imposed together with the Keto/Amino balance and the divisibility requirement is not introduced.

The stop-parameter equality as a two-axis condition. Instead of divisibility, one can require equal Δ values for Purine and Pyrimidine. Denote the differences of the service-codon parameters by $\Delta_{\text{stop}} = p_{\text{stop}} - n_{\text{stop}}$ and $\Delta_{\text{start}} = p_{\text{start}} - n_{\text{start}}$. Keto/Amino balance gives

$$\Delta_{\text{start}} = \Delta_{\text{stop}} + 1.$$

All service codons belong to Purine, and the difference between the Δ values of the two branches in the service-excluded pool is $\Delta_{\text{SE}}(\text{Pyrimidine}) - \Delta_{\text{SE}}(\text{Purine}) = 1$. The additional balance therefore requires

$$\Delta_{\text{start}} + 3\,\Delta_{\text{stop}} = 1.$$

Together these equations give $\Delta_{\text{stop}} = 0$ and $\Delta_{\text{start}} = 1$. Thus $p_{\text{stop}} = n_{\text{stop}}$ and $p_{\text{start}} - n_{\text{start}} = 1$ are obtained without requiring divisibility.

Together with Keto/Amino balance, this leaves the one-parameter family $(p_{\text{stop}}, n_{\text{stop}}) = (37+t, 37+t)$, $(p_{\text{start}}, n_{\text{start}}) = (1+t, t)$ for integer $t \geq 0$. Along this family the total neutron sum is $3700 + 4t$ and the total proton sum is $4292 + 4t$. The minimal total contribution is attained at $t = 0$, again giving Key 1. Both sums are divisible by 37 only when $t \equiv 0 \pmod{37}$; at the minimal assignment, divisibility is a consequence rather than a condition of this derivation. Equality of the stop parameters here depends on the agreement of the differences across two axes: $\delta P(\text{Trp}, \text{Ile}) - \delta N(\text{Trp}, \text{Ile}) = -1$ (Section 3.3.1) and the difference in Δ between Pyrimidine and Purine given above.

Why the balance in P and N is imposed on Keto/Amino. Equal proton and neutron sums for Purine and Pyrimidine would require different service-codon contributions:

$$p_{\text{start}} + 3\,p_{\text{stop}} = 212, \qquad n_{\text{start}} + 3\,n_{\text{stop}} = 211.$$

However, $212 \equiv -10 \pmod{37}$ and $211 \equiv -11 \pmod{37}$, whereas divisibility of the total sums requires residues 1 and 0, respectively. Such a balance is therefore incompatible with divisibility by 37. Among the three third-position axes, this requirement is compatible with balance in P and N on Keto/Amino and its structural copy on Strong/Weak. Balance in Δ on Purine/Pyrimidine does not impose these two separate equalities and is compatible with the assignment obtained.

The common basis of the derivations. Both derivations use equal assignments to the three stop codons, the same initial sums, and Keto/Amino balance. The additional condition differs: divisibility of the total sums by 37 or equality of Δ between Purine and Pyrimidine. With non-negativity, integrality, and minimality, both variants give Key 1. This is agreement between two systems of conditions with a common basis, rather than independent confirmations.

The resulting value $1850 = \frac{1}{2}T(\text{Octet I})$ allows the neutron assignments to be checked in reverse: $1850-1776 = 74 = 2\cdot37$ is assigned to the two stop codons in Amino, and $1850-1813 = 37$ to the stop codon and initiator in Keto. This again gives $n_{\text{stop}} = 37$ and $n_{\text{start}} = 0$. This check uses equality with half the Octet I sum after it has been obtained; it is not a third independent basis for the derivation.

3.5 Consequences under Key 1

Under Key 1, Keto/Amino balance extends to four 32-codon groups, while equality of the total neutron sum and the nucleon sum of Octet I connects several levels of partitioning. This section examines these consequences, the divisibility properties of the resulting sums, and the residual Purine/Pyrimidine asymmetry. An overview of the groups under Key 1 is given in Figure 4.

3.5.1 Alignment of the chemical-axis 32-codon groups

Under Key 1, Keto, Amino, Strong, and Weak have identical values of the four nucleon quantities:

$$T = 3996, \qquad P = 2146, \qquad N = 1850, \qquad \Delta = 296.$$

Keto/Amino balance ensures equality of the sums of all four groups in P and N, while the definitions $T = P + N$ and $\Delta = P - N$ ensure equality in T and Δ. Thus the four aligned groups are a consequence of the chosen assignment, rather than four separate confirmations. Their sums are half those of the complete table. All group values are given in `key1_nucleon-data.csv`.

Purine and Pyrimidine have the same Δ but differ in P, N, and T. This difference is examined in Section 3.5.5.

3.5.2 Macroscopic identities and propagation to chemical-axis halves

The relation between the complete table and the octets has three equivalent forms:

$$\begin{aligned} N_1(\text{All}) &= T(\text{Octet I}) = 3700 = 100 \cdot 37, \\ P_1(\text{All}) &= T_1(\text{Octet II}) = 4292 = 116 \cdot 37, \\ N_1(\text{Octet II}) &= P(\text{Octet I}) = 2000 = 2 \cdot 1000. \end{aligned}$$

This is one relation, obtained in Section 3.4.1 after determining the parameters. Subtracting $N(\text{Octet I}) = 1700$ from the first equality gives the third. Adding $P_1(\text{Octet II})$ to the third gives the second, since $T = P + N$. The transformations also hold in reverse. The sums of Octet I are independent of parametrization; those of Octet II include the service-codon assignments.

For each of the four aligned groups G—Keto, Amino, Strong, and Weak—this relation is reproduced at the level of its octet intersections:

$$\begin{aligned} N_1(G) &= T(G \cap \text{Octet I}) = 1850, \\ P_1(G) &= T_1(G \cap \text{Octet II}) = 2146, \\ N_1(G \cap \text{Octet II}) &= P(G \cap \text{Octet I}) = 1000. \end{aligned}$$

Each intersection contains 16 codons. In Octet I, each such group includes two codons from every four-fold degenerate family and therefore carries half the octet sums. In Octet II, equality of the halves along the two axes follows from complete-table balance after subtracting the equal contributions of Octet I. The equalities above are therefore half-scale forms of the same relation. This alignment of Octet II does not hold for Purine and Pyrimidine. The complete list of equalities is given in `key1_equalities.csv`.

Figure 4: Nucleon sums of codon groups under Key 1: each stop codon is assigned $(P, N) = (37, 37)$, and the initiator ATG is assigned $(P, N) = (1, 0)$. The group layout and divisibility color code are the same as in Figure 2; the comparison with Key 0 is shown in Figure 3. Generated with the accompanying visualizer `visualization.html`.

Rational profile. The complete table and the four aligned groups share the same ratios of nucleon quantities:

$$\frac{P}{T} = \frac{29}{54}, \qquad \frac{N}{T} = \frac{25}{54}, \qquad \frac{\Delta}{T} = \frac{2}{27}, \qquad P : N = 29 : 25.$$

This is a normalized expression of the same sums. For each of the four groups, $2146 = 29 \cdot 2 \cdot 37$ is both the proton sum of the group and the nucleon sum of its intersection with Octet II; $1850 = 25 \cdot 2 \cdot 37$ links the neutron sum to the intersection with Octet I. The ratio profiles of all groups are given in `key1_ratios.csv`.

3.5.3 Divisibility by 37 and by 999

Divisibility by 37. In the main derivation of Key 1, divisibility of the total sums in P and N is explicitly imposed. For neutrons it already holds at the minimal assignment satisfying balance alone; for protons it changes the minimum (Section 3.4.3). The resulting sums are

$$N_1(\mathrm{All}) = 100 \cdot 37, \qquad P_1(\mathrm{All}) = 116 \cdot 37.$$

Balance divides each sum into two equal integer parts. Since 37 is odd, both parts are also divisible by 37. Each of the four aligned groups has

$$T = 108 \cdot 37, \quad P = 58 \cdot 37, \quad N = 50 \cdot 37, \quad \Delta = 8 \cdot 37.$$

Divisibility of T and Δ follows by adding and subtracting P and N; in particular, $\Delta_1(\mathrm{All}) = 4292 - 3700 = 592 = 16 \cdot 37$.

A more general property holds for the nucleon sum:

$$T_1(G) \equiv T_0(G) \pmod{37}$$

for any codon group G. In the transition from Key 0 to Key 1, the contribution of each stop codon to T increases by $74 = 2 \cdot 37$, whereas that of ATG changes by $1 - 149 = -148 = -4 \cdot 37$. Divisibility of T by 37 is therefore preserved in this transition regardless of the group's composition. The divisibility records under Key 1 are given in `key1_divisibility-37.csv`.

Divisibility by 999. The total nucleon sum has the form

$$T_1(\mathrm{All}) = 7992 = 8 \cdot 999 = 6^3 \cdot 37.$$

Unlike divisibility by 37, divisibility by $999 = 27 \cdot 37$ was not imposed as a separate condition. It characterizes the minimal solution obtained. Each of the four aligned halves has $T = 3996 = 4 \cdot 999$; this follows by halving the total sum and is not another numerical coincidence.

3.5.4 The Δ-invariant across all six 32-codon groups

Under Key 1, all six third-position 32-codon groups have the same proton-neutron difference:

$$\begin{aligned}\Delta_1(\mathrm{Keto}) = \Delta_1(\mathrm{Amino}) = \Delta_1(\mathrm{Strong}) &= \Delta_1(\mathrm{Weak}) \\ &= \Delta_1(\mathrm{Purine}) = \Delta(\mathrm{Pyrimidine}) = 296.\end{aligned}$$

For the first four groups, equality follows from balance in P and N. Equality of Purine and Pyrimidine in Δ is not imposed separately in the main derivation; in the alternative two-axis derivation it serves as a condition in place of divisibility (Section 3.4.3). Its role therefore differs between the two derivations.

The common difference is reproduced at lower levels of partitioning. Every eight-codon group of Octet I selected by a single nucleotide in the third position has $\Delta = 75$ independently of parametrization (Section 3.1.2). Under Key 1, every corresponding group of Octet II has $\Delta = 73$:

$$\Delta(G_X \cap \text{Octet I}) = 75, \qquad \Delta_1(G_X \cap \text{Octet II}) = 73 \qquad (|X| = 1).$$

In the pyrimidine groups, 73 is also independent of parametrization (Section 3.1.3); in the purine groups it is attained under the derived assignment. Adding two eight-codon groups within each octet gives

$$\Delta(G_X \cap \text{Octet I}) = 150, \qquad \Delta_1(G_X \cap \text{Octet II}) = 146 \qquad (|X| = 2).$$

Adding across the octets gives, for each one-nucleotide group of the complete table,

$$\Delta_1(G_X) = 75 + 73 = 148 = 4 \cdot 37 \qquad (|X| = 1).$$

Under Key 0, the {G} group had $\Delta_0(\{\text{G}\}) = 158$, whereas the other three already had 148 (Section 3.2.3). Under Key 1, all four values coincide. Both routes of addition reach the 32-codon level:

$$296 = 2 \cdot 148 = 150 + 146.$$

Thus the cascade describes one agreement at several levels. It also persists for nonminimal assignments in the two-axis family of Section 3.4.3: increasing both parameters by the same amount leaves Δ unchanged for every service codon.

3.5.5 Purine/Pyrimidine asymmetry

Under Key 1, the Purine/Pyrimidine axis retains the following differences between the sums of its halves:

$$\begin{aligned} N(\text{Pyrimidine}) - N_1(\text{Purine}) &= 100, \\ P(\text{Pyrimidine}) - P_1(\text{Purine}) &= 100, \\ T(\text{Pyrimidine}) - T_1(\text{Purine}) &= 200, \\ \Delta(\text{Pyrimidine}) - \Delta_1(\text{Purine}) &= 0. \end{aligned}$$

The entire asymmetry lies in Octet II: in Octet I, the purine and pyrimidine halves have equal sums because its families are four-fold degenerate.

In the service-excluded pool, the Pyrimidine–Purine differences are 212 in P and 211 in N (Section 3.1.1). Key 1 adds 112 protons and 111 neutrons to the purine half, so

$$212 - 112 = 211 - 111 = 100.$$

Equality of the two shifts is equivalent to equality of Δ between Purine and Pyrimidine. Their common magnitude of 100 is obtained from the initial sums after assigning the parameters; it was not imposed as a condition of the derivation.

Equalities at the one-nucleotide level. Equality of the sums of the {C} and {T} groups is independent of parametrization and follows from pyrimidine synonymy (Section 2.4). Since Keto combines {G} and {T}, whereas Amino combines {A} and {C}, balance in P and N gives

$$P_1(\{\text{G}\}) = P_1(\{\text{A}\}), \qquad N_1(\{\text{G}\}) = N_1(\{\text{A}\}).$$

This restates balance and holds for any assignment that satisfies it. Under Key 1, the two pairs of groups have the values

$$\begin{aligned} \big(T(\{\text{C}\}), P(\{\text{C}\}), N(\{\text{C}\})\big) &= (2048, 1098, 950), \\ \big(T_1(\{\text{G}\}), P_1(\{\text{G}\}), N_1(\{\text{G}\})\big) &= (1948, 1048, 900), \end{aligned}$$

with the same values for {T} and {A}, respectively. The difference between the pairs is 50 in P and N and 100 in T; doubling reproduces the differences between the two 32-codon halves. Subtracting the equal Octet I contributions leaves $T = 1023$, $P = 548$, $N = 475$ in each of the groups $\{G\} \cap$ Octet II and $\{A\} \cap$ Octet II under Key 1.

Inside Octet II at the 16-codon level. The purine and pyrimidine halves of Octet II have neutron sums of 950 and 1050 and proton sums of 1096 and 1196, respectively. They deviate by ± 50 from the mean values $N = 1000$ and $P = 1146$ of the halves of this octet along the Keto/Amino and Strong/Weak axes. This is another expression of the same asymmetry: the two halves lie symmetrically about the mean but are not equal to each other.

The limit of alignment. The residual asymmetry does not imply that complete balance across the three axes is impossible. In the two-axis family of Section 3.4.3, where the stop codons have parameters $(37 + t, 37 + t)$ and the initiator has $(1 + t, t)$, both Pyrimidine–Purine shifts equal $100 - 4t$. At $t = 25$, the assignments $(62, 62)$ to the stop codons and $(26, 25)$ to the initiator equalize all six halves in P and N. However, the total sums become $P = 4392$ and $N = 3800$ and are not divisible by 37; the neutron equality with $T(\text{Octet I}) = 3700$ is also lost. Key 1 fixes the minimal reconciliation of balance with divisibility, retaining this asymmetry.

3.6 Synthesis

The observed regularities connect two levels of organization of the standard genetic code: the structure of codon groups and the nucleon composition of the encoded amino acids. The partitions are based on the chemical properties of bases, pyrimidine synonymy, and Rumer's octets. In the adopted representation of free neutral amino acids, proton and neutron counts are uniquely determined by molecular composition, while group sums account for the number of codons encoding each amino acid. The choice of the third position, molecular representation, and modulus 37 defines the analytical framework; these choices alone do not entail the observed numerical values.

The structural partitions exhibit arithmetic agreement even before values are assigned to service codons. In Octet I, the proton and neutron sums are 2000 and 1700, and their sum, $3700 = 100 \cdot 37$, connects a decimal scale with modulus 37. The octet's structure ensures exact scaling of these quantities across its groups but does not determine their initial values. For the entire service-excluded pool, the neutron sum is $3589 = 97 \cdot 37$, while the proton sum is $4180 = 113 \cdot 37 - 1$ (Section 3.1).

The local asymmetry agrees with this global pattern. The difference between Keto and Amino reduces to the tryptophan–isoleucine pair, whose differences are 37 in neutrons and 36 in protons. Among the 190 pairs of canonical amino acids, this combination of differences occurs only for tryptophan–isoleucine and tryptophan–leucine. The agreement is explained by isoleucine and leucine being isomers with identical nucleon counts (Section 3.3.1). The even proton counts of all canonical amino acids rule out a proton difference of 37 but do not determine how far the actual difference must deviate from it. The value 36 is the nearest permitted value below 37, and the proton and neutron imbalances differ by exactly one unit (Section 3.3.2).

One-unit deviations occur at different levels of the structure and have different arithmetic origins. In Octet I, the values $75 = 2 \cdot 37 + 1$ arise by dividing its difference $\Delta = 300$ among four equal eight-codon groups. In the pyrimidine groups of Octet II, the value is displaced in the opposite direction: $73 = 2 \cdot 37 - 1$. When the octets are combined, the deviations cancel: $75 + 73 = 148 = 4 \cdot 37$. In the service-excluded pool, the {A} branch of Octet II also has $\Delta = 73$, an equality that does not follow from pyrimidine synonymy. Only the {G} branch differs, with a value of 72; the excluded ATG position lies in this branch (Section 3.2.3). A further agreement appears under Key 0: methionine's nucleon count, $149 = 4 \cdot 37 + 1$, compensates for the remainder of -1 in the pool's nucleon sum. Thus, one-unit deviations have several sources, and their recurrences propagate through structural relations among groups.

The arrangement of service codons allows local balance to be reconciled with global sums. Keto contains one stop codon and ATG, whereas Amino contains two stop codons. With the same assignment to all stop codons, the minimal non-negative integer compensation for the neutron imbalance assigns 37 neutrons to each stop codon and zero to ATG. This gives

$$N_1(\text{All}) = 3589 + 3 \cdot 37 = 3700 = T(\text{Octet I}).$$

The coincidence with the Octet I sum was not used to select the compensation. Here, the magnitude of the local imbalance, the number and arrangement of stop codons, and the sum of structural Octet I agree.

On the proton side, moving from the minimal balanced assignment to the minimal balanced assignment with divisibility by 37 requires four additional protons— one for each service position. Their equal distribution between Keto and Amino preserves balance. Since all four positions belong to Purine, the same increment eliminates the difference in Δ between Purine and Pyrimidine while preserving the original pyrimidine sums. The resulting assignment is Key 1: $(P, N) = (37, 37)$ for each stop codon and $(1, 0)$ for ATG (Section 3.4). Its simplicity expresses the compatibility of the initial numbers with the arrangement of service positions; uniqueness is relative to the adopted conditions and minimality criterion.

The resulting assignment brings these relations together in the complete table. Several macroscopic equalities express a single relation with Octet I; equality of Δ is compatible with Pyrimidine exceeding Purine by 100 in both proton and neutron sums. The nucleon sum $7992 = 8 \cdot 999$ adds to the numerical profile, although divisibility by 999 was not imposed as a separate condition (Section 3.5). The result concerns the agreement among molecular counts, structural partitions, and formal assignments; repeated manifestations of one relation do not constitute separate pieces of evidence. The specificity of this agreement is assessed next through comparison with alternative codes, deterministic controls, and a null model (Section 4).

4 Discussion

This section examines how far the regularities reported above are specific to the standard code and its structure, rather than artifacts of the analysis. Section 4.1 compares the standard code against the alternative NCBI codes on the sense-pool arithmetic, testing both the global alignment with the 37-lattice and the robustness of the Trp–Ile localization; Section 4.2 asks whether the third codon position and the modulus 37 are distinguished from other positions and divisors; and Section 4.3 assesses the concentration on the 37-lattice against a random-code null. Section 4.4 situates the results relative to prior work, and Section 4.5 states the limitations and outlook.

4.1 Specificity across alternative genetic codes

The standard genetic code is one of 27 translation tables in the NCBI registry. The parametrization-independent sense-pool arithmetic of Section 3.1 admits direct comparative tests across these tables: for each alternative code one can ask whether the same arithmetic holds. Two such tests are reported here. The first concerns the global alignment of the sense pool with the lattice of step 37—the neutron deficit and the proton residue of Section 3.1.1. The second concerns the localization of the Keto/Amino asymmetry in the single pair Trp–Ile (Section 3.3.1). Both quantities are properties of the sense pool and require no assignment to the service codons; the closure of the first pair under Key 1 is a separate matter, treated in Section 3.4. The two tests probe different aspects of the arithmetic and, as shown below, identify different sets of matching alternative codes.

Because both tests depend on which codons are counted as stops, both are computed under two model choices for the sense pool:

- Sense pool, Model 1 (primary): all codons listed as stops in the NCBI registry are excluded, even when they are read as amino acids in context-dependent codes. The primary start codon ATG is excluded as the universal initiator. Initiation is a property of the first codon position rather than of the triplet itself, so exactly one initiator codon is removed; alternative start codons such as GTG and TTG carry their elongation meaning at every internal position and are counted as sense codons.

- Sense pool, Model 2 (alternative): in the three codes with context-dependent stops (Tables 27, 28, 31), codons that function as amino acids in mid-sequence and as stops at gene termini are treated as sense codons. For all other tables, Model 2 coincides with Model 1.

The model conventions are documented in the datasets accompanying `deficit_models_analysis.csv` and `keto_amino_balance_models.csv`.

Global alignment of the sense pool with the lattice. The first test records two quantities of the sense pool: the neutron deficit relative to a reference total, $\Delta_N = T_{\mathrm{ref}} - N(\mathcal{C}_{\mathrm{SE}})$, and the proton residue $P(\mathcal{C}_{\mathrm{SE}}) \bmod 37$. For the standard code these are $\Delta_N = T(\text{Octet I}) - N(\mathcal{C}_{\mathrm{SE}}) = 3700 - 3589 = 111 = 3 \cdot 37$ and $P(\mathcal{C}_{\mathrm{SE}}) \equiv -1 \pmod{37}$ (Section 3.1.1); under Key 1 the first deficit is closed by the three stop codons to align $N(\text{All})$ with the invariant $T(\text{Octet I})$, and the proton residue by the start-codon parameter $p_{\text{start}} = 1$. For the neutron deficit, three model choices of T_{ref} are considered: the constant $T(\text{Octet I}) = 3700$ of the standard code, the total nucleon content of the eight structural Octet I families under the amino-acid assignments of the specific table, and the total nucleon content of all 4-fold degenerate codon families in the specific table. For the proton residue the relevant quantity is intrinsic to the sense pool and requires no reference total. The full set of computed values across all tables and model combinations is tabulated in `deficit_models_analysis.csv`.

The two records give parallel outcomes: in each, the arithmetic property holds in exactly the same three tables. Table 1 (the standard code) and Table 11 (the Bacterial, Archaeal and Plant Plastid Code) have $\Delta_N = 111 = 3 \cdot 37$ under all three reference models and $P(\mathcal{C}_{\mathrm{SE}}) \equiv -1 \pmod{37}$ under both sense-pool models; their amino acid assignments are identical and they share the same set of stop codons, so arithmetically they are the same code applied in different domains. Table 28 (the Condylostoma Nuclear Code) gives the same values as the standard code under Model 1 and different values under Model 2: its three context-dependent stop codons, once excluded alongside ATG, leave a sense codon set identical to the one used for the standard code, whereas under Model 2 they are returned to the pool and neither the deficit nor the residue matches. In the remaining 24 tables neither property holds under any model combination. That the two records identify the same tables reflects a single underlying sense pool rather than independent agreement: Tables 1, 11, and (under Model 1) 28 all share the standard code's sense-codon assignments.

Across the 27 tables, six reasonable definitions of the neutron deficit and two of the proton residue, only the standard code and its exact codon-to-amino-acid equivalent (Table 11) satisfy both conditions under every model. Tables that reassign codons within the structural Octet I families, shift the set of 4-fold-degenerate families, or use a different number of stop codons destroy the match between the sense pool and the lattice of step 37. The global alignment is therefore not a generic feature of genetic codes.

Preservation of the Trp–Ile localization. The second test asks whether the localization of the Keto/Amino asymmetry in the pair Trp–Ile survives in alternative codes. The relevant signature is the pair of sense-pool differences $N_{\mathrm{SE}}(\text{Keto}) - N_{\mathrm{SE}}(\text{Amino}) = 37$ and $P_{\mathrm{SE}}(\text{Keto}) - P_{\mathrm{SE}}(\text{Amino}) = 36$, the $(37, 36)$ signature; the values across all tables are computed in `keto_amino_balance_models.csv`. The signature is reproduced exactly in seven tables. Five—Table 1 (Standard), Table 6 (Ciliate, Dasycladacean and Hexamita Nuclear), Table 11 (Bacterial, Archaeal and Plant Plastid), Table 29 (Mesodinium Nuclear), and Table 30 (Peritrich Nuclear)—reproduce it under both sense-pool models; the remaining two, Table 27 (Karyorelict Nuclear) and Table 28 (Condylostoma Nuclear), reproduce it only under the primary interpretation of context-dependent stop codons.

The mechanism is structural and operates differently in different tables. Table 11 has an amino acid assignment identical to the standard code and shares its set of stop codons, so the signature holds trivially. In Tables 6, 29, and 30 the codons TAG and TAA are reassigned to the same amino acid (Gln, Tyr, or Glu respectively); since TAG sits in the Keto branch of family TA and TAA in the Amino branch, this symmetric reassignment contributes identically to both branches of the sense pool and the Keto/Amino differences are preserved. No codons other than TAA and TAG are reassigned in these tables; in particular, neither TGG (Trp) nor any codon of Ile is changed, and the Trp/Ile localization of Section 3.3.1 continues to hold. Tables 27 and 28 involve context-dependent stop codons and behave differently under the two models: Table 28 declares all three of TAA, TAG, TGA as context-dependent stops, so under Model 1 its sense pool is identical to that of the standard code and the signature is reproduced, while under Model 2 the three codons re-enter the pool and it is lost; Table 27 permanently reassigns TAA and TAG to Gln (the same symmetric mechanism) and treats only TGA as a context-dependent stop, so Model 1 preserves the signature while Model 2, including TGA as Trp in the Keto branch, breaks it. Tables that reassign ATA (typically to Met in mitochondrial codes) or TGA (typically from stop to Trp) change the amino acid content of the carrier codons and no longer reproduce the signature.

The asymmetry between nuclear and mitochondrial codes is informative. All seven tables matching the $(37, 36)$ signature are nuclear or bacterial; none of the thirteen mitochondrial translation tables in the NCBI registry reproduces it, the failure tracing in each case to the reassignment of ATA or TGA, both altering codons on which the localization depends.

Relation between the two tests. A finer decomposition connects the seven matching tables to the first test. In the standard code the four sense-pool half-quantities have the residues $(N_{\text{Keto}}, N_{\text{Amino}}, P_{\text{Keto}}, P_{\text{Amino}}) \equiv (0, 0, -1, 0) \pmod{37}$, an arithmetic profile combining divisibility of each half-pool by 37 with a one-unit proton residue: parity bars the proton half-pools from the odd multiples of 37, and the specific nucleon counts place the residue at exactly one. Tables 1, 11, and 28 (under Model 1) reproduce all four residues exactly—for Table 11 by identity with the standard code, for Table 28 under Model 1 because its sense pool coincides with the standard once the context-dependent stops are excluded—and these are exactly the three tables that match in the global-alignment test above. The four additional tables matching the $(37, 36)$ signature (6, 27, 29, 30) show a different profile: N_{Keto} and N_{Amino} share the same non-zero residue mod 37, and P_{Keto} and P_{Amino} differ by one unit, but the absolute level has detached from the lattice. This is the signature of symmetric reassignment of TAA and TAG to a common amino acid: each half-pool acquires the same non-zero contribution, so the Keto/Amino differences are preserved while the divisibility of each half by 37 is lost. Among the 27 tables, 13 preserve the divisibility of at least one half-pool by 37, three of them mitochondrial (Tables 4, 16, and 23).

The global-alignment test measures the absolute alignment of the whole sense pool with the 37-lattice and matches only tables arithmetically equivalent to the standard code in their sense pool (Tables 1, 11, and 28 under Model 1). The localization test measures the preservation of a relational pattern between the Keto and Amino branches and additionally matches tables in which the only divergence from the standard code is a symmetric reassignment of TAA and TAG (Tables 6, 27, 29, 30). Together they show that the parametrization-independent sense-pool arithmetic of the standard code, in both its absolute and its relational form, is specific to the standard code among the known natural codes, holding elsewhere only in tables that are arithmetically equivalent to it or differ from it by a symmetric reassignment of TAA and TAG.

4.2 Specificity of the third codon position, the modulus 37, and the molecular representation

Two features are built into the framework of this work rather than derived from the data: the three chemical axes are read off the third codon position (Section 2.3), and divisibility by 37 is the primary criterion (Section 2.6). Two deterministic controls test whether the observed structure is specific to these two choices or would arise generically.

Codon position. Applying the same three chemical axes to the first and second codon position, instead of the third, removes the structure almost entirely. At the third position the Amino and Weak groups have all four nucleon quantities divisible by 37, the Keto and Strong groups have their neutron pool divisible, and the Pyrimidine group has Δ divisible. At the first position the only divisibility is the neutron pool of the Keto/Amino axis, present in both its complementary halves, and this is not an independent fact: Keto and Amino are complementary, so their sense-pool neutron counts sum to $N(\mathcal{C}_{\mathrm{SE}}) = 3589 = 97 \cdot 37$ (Section 3.4.1), and divisibility of one half forces divisibility of the other. At the second position no nucleon quantity of any axis is divisible by 37. The control thus shows that divisibility by 37 along the chemical axes is confined to the third codon position: the same axes applied to the first and second positions do not meet the 37-lattice, the single first-position exception being the dependent one noted above. This excludes the alternative in which any equal-size split of the sense pool into chemical halves would align with the 37-lattice irrespective of position. The full values for all three positions are tabulated in `position_axis_analysis.csv`.

Choice of modulus. For every prime up to 97 we count how many of the four nucleon quantities of the codon groups it divides, over three value sets: the parametrization-independent groups (those with no service codon, on which 37 cannot be imposed), the groups under Key 0 (which does not impose 37), and the groups under Key 1 (whose derivation does). The scan ranges over primes rather than all integers because divisibility by a composite is the conjunction of divisibilities by its prime-power factors and introduces no new modulus, and because raising a prime to a higher power lowers the chance rate $1/m$ and thereby inflates the excess over the same quantities without adding information; the prime is the appropriate unit. The informative statistic is accordingly the excess over the chance rate $1/p$, computed over distinct values so that quantities forced to coincide by the structure of the code are not counted more than once.

Three primes divide their share at essentially the chance rate (3, 7 and 13). Two exceed it modestly, by less than a factor of three across the three value sets: 2, because every canonical amino acid has an even proton count (Section 3.3.2), and 5, reflecting the round-number regularity already reported in Section 3.6, which the scan thus corroborates. Against this background 37 divides five of the twenty-eight distinct parametrization-independent quantities, thirteen of the seventy distinct Key 0 quantities, and ten of the forty-seven under Key 1 — in each case several times the naive rate 1/37. This excess ratio is descriptive, not a p-value: the scan is deterministic and has no null model, and the nested group hierarchy makes the quantities non-independent, so the ratio alone cannot be read as a significance. An additional descriptive distinction not captured by that ratio is that the multiples of 37 contain several distinct odd cores, whereas the comparable factors 73 and 61 each reduce to a single core. Because the effect is already present in the value sets where 37 is not imposed, it is in any case not an artifact of the Key 1 derivation.

Three features set 37 apart from the two small primes, all independent of the number base. It is a larger prime, so each divisible quantity is rarer under the null. It has no structural counterpart to proton parity. And, unlike 5 — whose multiples in these value sets are almost all multiples of 25, so that the larger excess one would attribute to 25, 50 or 100 only re-scores the same quantities against a smaller chance rate — the modulus 37 occurs at a single power: no quantity is divisible by 37^2, so its excess does not grow when the modulus is enlarged. Reducing each quantity to its odd part — dividing out the powers of two that the nested group hierarchy

(Octet I and its sixteen- and eight-codon sub-groups) introduces mechanically, since a group of size $2k$ has twice the nucleon counts of its size-k refinement — sharpens this. Among the Key 0 quantities, 37 divides ten distinct odd cores; the multiples of $73 = 2 \cdot 37 - 1$ (the value of Δ for the Octet II eight-codon groups) collapse to that single core, and the multiples of 61 to the single core $549 = 9 \cdot 61$, to which the pyrimidine proton pool $P(\text{Pyrimidine}) = 2196 = 2^2 \cdot 3^2 \cdot 61 = 4 \cdot 549$ and the proton count of each pyrimidine letter $1098 = 2 \cdot 549$ both reduce. The factor 73 is thus an echo of the modulus itself; the factor 61 is an isolated factorization unrelated to 37 (its quantities deviate from the nearest multiples of 37 by $+13$ and -12, not by ± 1 or ± 2); and the factor 5 reduces entirely to 10 and 25, every quantity here divisible by 5 being divisible by 10 or by 25. The odd-core counts are tabulated in the corresponding column of `prime_divisibility_scan.csv`.

Representation dependence and the invariant relational core. The nucleon counts throughout are those of the free (neutral) amino acids — the canonical molecular identity of the species each codon encodes — under the most-abundant-isotope convention (^{1}H, ^{12}C, ^{14}N, ^{16}O, ^{32}S). Any alternative representation of the encoded amino acid differs from this one by a fixed nucleon offset per residue: the peptide residue removes one water (10 protons, 8 neutrons), the zwitterion removes nothing. Under such a uniform offset (s_P, s_N), a linear combination of the counts is invariant if and only if its signed codon coefficients sum to zero; the control `representation_sensitivity.csv` records every anchor under all three representations. This splits the regularities in two. The relational regularities — the neutron and proton differences of the localizing pair Trp–Ile ($|\delta N| = 37$, $|\delta P| = 36$) and the Keto/Amino and Strong/Weak axis differences (37, 36) — are differences between codon sets of equal cardinality and are therefore invariant under every such re-representation. The absolute regularities — the sense-pool sums $N(\mathcal{C}_{\text{SE}}) = 97 \cdot 37$ and $P(\mathcal{C}_{\text{SE}}) \equiv -1 \pmod{37}$, the deficit $111 = 3 \cdot 37$, and the divisibility of the individual axis half-pools — are not: a uniform offset preserves the neutron alignment only when $60\, s_N \equiv 0 \pmod{37}$, i.e., $s_N \equiv 0$ (since $\gcd(60, 37) = 1$), which among the representations considered excludes the peptide residue, while the neutral and zwitterionic forms of the free amino acid have the same elemental composition and therefore the same nucleon counts; under the peptide residue the sums become $N = 3109 \equiv 1$ and $P = 3580 \equiv 28 \pmod{37}$. The absolute alignment is therefore a property of the standard code's assignment in the canonical free-amino-acid representation, fixed a priori by molecular convention rather than by tuning, while the relational structure — the load-bearing part of the analysis — is representation-independent.

The functional-group asymmetry across positions. One observation on the Keto/Amino axis reaches beyond the third position. The asymmetry $N(\text{Keto}) - N(\text{Amino})$ that localizes to the Trp–Ile pair at the third position (Section 4.1) equals 37 there; the same Keto/Amino axis — the $\{G, T\}$ versus $\{A, C\}$ split — yields a quantity divisible by 37 at the first position as well (N-asymmetry $-333 = -9 \cdot 37$) and at the second (T-asymmetry $185 = 5 \cdot 37$). The divisible quantity is N, T, N by position, fixed after inspection rather than in advance, so this is a single observation along one axis rather than a control of the kind above. The three are arithmetically independent: each is a sum over a different partition of the same sixty codons, and the divisibility at one position neither follows from nor implies it at another; what they share is only the choice of axis. Across the 27 NCBI translation tables, the result depends on the sense-pool model. Under Model 1, the full three-position signature occurs for Tables 1, 11, and 28, whereas under Model 2 it occurs only for Tables 1 and 11. Table 28 is not an independent replication under Model 1: after its three context-dependent stop codons are excluded, its analyzed pool is identical to that of the standard code. Thus, in neither model is the full signature reproduced by a table with a non-equivalent analyzed pool. Apart from these full matches, no table reaches more than one position. In particular, Tables 6, 29, and 30 reach only the third position under both models, whereas Table 27 reaches only the third position and only under Model 1. Three caveats further qualify it. Unlike the third-position case, the first- and second-position asymmetries do not localize to a single amino-acid pair — each receives contributions from nineteen of the twenty amino acids, because the cancellation that isolates Trp and Ile relies on third-position synonymy, which the other positions lack. Their non-divisible components carry no analogue of the parity-induced residue -1 that accompanies the third-position value. And, as in Section 4.1, the 27 tables are not statistically independent, so this is evidence of specificity among natural codes rather than of improbability under a null model. The per-table values are tabulated in `position_asymmetry_ncbi.csv`.

All four controls are deterministic, use exact integer arithmetic, and are regenerated by `controls.py` (Section 2.7).

4.3 Significance under a random-code null

The controls of Section 4.2 show that divisibilities concentrate in third-position groupings and that modulus 37 stands out among the tested prime moduli; the comparison of Section 4.1 shows that, among the 27 NCBI translation tables, the tested combination occurs only in the standard code and its exact codon-to-amino-acid equivalent. Specificity among realized codes, however, is not improbability under an explicit null. We therefore test the standard code against a random-code null of the kind used in studies of genetic-code optimality [2]: the architecture of the code is held fixed (the 64 codons, the synonymous blocks with their sizes, the three stop codons, the Rumer partition, and the third-position axes), and the only thing randomized is which amino acid, carrying its fixed proton and neutron counts, occupies each of the twenty blocks. A trial is a uniformly random permutation of the twenty real paired proton and neutron profiles over the twenty synonymous blocks. The modulus 37 is taken a priori from the prior literature; the statistical summaries were formally defined before the final Monte Carlo run. Because the analysis was not preregistered before inspection of the observed structure, the results are interpreted as exploratory.

S1–S3 and S4sense are evaluated over the sixty-codon sense pool without assigning any values to service codons. S4sense averages the distances to the nearest multiple of 37 for the four quantities T, P, N, and Δ over all 33 groups, for 132 distances in total. Under S4a the stop codons contribute zero, while the contribution at ATG in each random code is the profile of the amino acid assigned by the permutation to the ATG block. Under S4b each random code receives its own minimal non-negative Key 1, re-derived by the same Keto/Amino balance and global divisibility conditions used for the standard code.

For every statistic the Monte Carlo tail probability is calculated as $p_{\mathrm{MC}} = (b+1)/(M+1)$, where b is the number of random codes at least as extreme as the standard code and $M = 10^6$. A 95% Monte Carlo uncertainty interval is reported with each estimate. The computation is reported in `mc_significance.py` and summarized in `mc_significance.csv`. An independent implementation, using a different construction of the codon groups, a direct search for Key 1, and a different random seed, reproduces the observed statistics and tail probabilities within the expected Monte Carlo error (`verify_mc.py`).

Six statistical summaries are considered. S1 tests the joint divisibility of two anchors: $N(\mathcal{C}_{\mathrm{SE}}) = 3589 = 97 \cdot 37$ and $T(\text{Octet I}) = 3700 = 100 \cdot 37$. Both events occur in a fraction 0.000800 of random codes (Monte Carlo 95% interval 0.000746–0.000856). This is the directly estimated joint probability; no assumption that the two events are independent is used in its calculation.

S2 counts divisibility by 37 among the four quantities T, P, N, and Δ for nine sense-pool groups: the full pool, the six chemical-axis half-pools, and the two octets. The standard code has 13 divisible quantities among 36 nominal entries, against a null mean of 1.0299. A value of 13 or greater occurred in 39 of 10^6 trials, giving $p_{\mathrm{MC}} = 4.0 \times 10^{-5}$ with a 95% interval of 2.9×10^{-5}–5.3×10^{-5}. The components of the count are dependent: $T = P + N$, $\Delta = P - N$, and the Keto/Strong and Amino/Weak groups form exact twin pairs. These dependencies are reproduced in every permutation and are therefore already included in the null distribution of the composite statistic.

S3 applies the same count to the 17 groups that contain no service codon at all, giving 68 nominal entries. The standard code has 14 divisible quantities against a null mean of 2.1311, $p_{\mathrm{MC}} = 0.025135$, and a 95% interval of 0.024829–0.025443. Eleven of the fourteen divisibilities are structural copies of $T(\text{Octet I}) = 100 \cdot 37$, while the remaining three form the pyrimidine Δ-cascade ($\Delta(\text{Pyrimidine}) = 8 \cdot 37$ and $\Delta(\{\text{C}\}) = \Delta(\{\text{T}\}) = 4 \cdot 37$). S3 therefore records the propagation of two structural cascades and is treated as a secondary summary.

S4sense measures not only exact divisibility but the overall proximity of the sense pool to the lattice of step 37. For each of the 33 groups, the distances of T, P, N, and Δ to the nearest multiple of 37 are averaged, with zero included as a distance of zero; 132 quantities are included in total. The standard code has mean distance 4.8182 against a null mean of 9.1495. An equal or smaller distance occurs in a fraction 0.000274 of random codes, with a 95% interval of 0.000242–0.000307. This statistic assigns no values to service codons. Because the summary was defined after inspection of the observed structure, it is treated as an exploratory continuous sensitivity analysis.

S4a applies the same distance under Key 0. In the standard code ATG encodes methionine; in each random code its contribution is the amino acid actually assigned by the permutation to the ATG block, while the stop codons contribute zero. The standard code has mean distance 5.5606 against a null mean of 9.0157, $p_{\mathrm{MC}} = 0.002807$, and a 95% interval of 0.002704–0.002912. The concentration is therefore already present under the naive Key 0, although Key 0 remains a formal assignment of service-codon values.

S4b tests the Key 1 derivation symmetrically: the standard code and every random code are evaluated under their own minimal non-negative key, derived from the Keto/Amino balance and global divisibility by 37. The fixation of this axis and the compatibility of the conditions are established in Section 3.4.3. The standard code has mean distance 4.6364 against a null mean of 7.7397, $p_{\mathrm{MC}} = 0.021106$, and a 95% interval of 0.020825–0.021389. Thus the standard code's Key 1 is moderately atypical among keys re-derived for random codes.

The two most sensitive sense-pool summaries — the discrete S2 and the continuous S4sense — describe the same overall concentration from different directions. Their p-values are dependent and must not be multiplied or interpreted as two independent lines of evidence. S1 records two individual anchors of the structure, whereas the weaker S3 shows that restriction to groups with no service positions leaves mainly the Octet I and pyrimidine-Δ cascades.

S4a and S4b address separate questions about the two formal assignments to service codons. S4a shows that the concentration is retained under the naive Key 0. S4b evaluates the entire Key 1 derivation rather than the fixed values of the standard key and gives a more moderate tail probability. The assignment-free sense-pool results therefore form the central statistical part of the analysis, while S4a and S4b are secondary tests of its extension to the complete table.

All reported p_{MC} values are nominal tail probabilities for the individual statistics; no multiplicity correction across the six statistics was applied.

The null model fixes the standard code's degeneracy architecture and the real paired proton and neutron counts of the amino acids and randomizes only their assignment to blocks. A different generative model would give different probabilities. The modulus 37, the Rumer partition, and the chemical axes have substantive a priori origins, but the choice of global statistical summaries was not preregistered before inspection of the structure. The results therefore characterize the exploratory atypicality of the standard code under one explicit combinatorial model rather than confirming a preregistered hypothesis. Improbability under this null concerns combinatorial codes, a different question from specificity among the 27 natural codes in Section 4.1; it speaks to no biological mechanism.

4.4 Relation to prior work

Three prior contributions are particularly relevant to the arithmetic structure analyzed in this work. Yu. B. Rumer [6, 7] introduced the partition of the 64 codons of the standard code into two equal classes: Octet I consisting of the eight fully degenerate families, and Octet II consisting of the eight heterogeneous families. The two classes are related by the involution $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$, under which each family of one class is mapped to a family of the other. Shcherbak and Makukov [8] observed that the number 37 plays a recurrent role in the genetic code, appearing as a divisor of amino acid nucleon sums across various groupings of codons. Panov and Filatov [5] revisited and extended this analysis. They introduced the per-codon counting rule, in which each of the 64 codons contributes independently the nucleon count of the amino acid it encodes. They also proposed that stop codons be assigned a total nucleon count of 74 each rather than zero.

The present work differs from these in several methodological choices. Shcherbak and Makukov, and Panov and Filatov following them, decompose each amino acid into a constant part with nucleon count 74 and a variable side chain, and many of their nucleon sums are formed over side chains alone. This decomposition forces a special treatment of proline, whose cyclic side chain reduces the nucleon count of its constant part to 73 rather than 74: to bring it in line with the other nineteen amino acids, Shcherbak and Makukov introduce a virtual transfer of one nucleon from its side chain to its constant part, an operation they call the *activation key*. The present work uses full amino acid nucleon counts without separation into constant and side-chain components, and the question of how to treat proline does not arise. Shcherbak and Makukov also adopt a counting rule in which each amino acid contributes once per codon family rather than once per codon; Panov and Filatov replace this with the per-codon rule, which the present work also uses. Finally, all nucleon sums in the prior works are formed over the total nucleon count T (or over its restriction to side chains); the separation of T into protons P and neutrons N, and the analysis of the difference $\Delta = P - N$, are introduced in the present work and underlie a class of arguments unavailable when only T is considered.

The two prior works also differ from ours in the systematics by which codons are grouped. They apply several different grouping schemes in parallel (Gamow's sorting and others), each tailored to a particular set of patterns. The present work formulates a single grouping system based on the Rumer octets and the third-position chemical axes, and analyzes the arithmetic structure within it consistently. The parametrization of service codons also differs: Shcherbak and Makukov treat stop codons as contributing zero and identify ATG with Met throughout, Panov and Filatov propose in addition the assignment of total nucleon count 74 to stop codons as an alternative, while the present work considers the parametrization of service codons as part of the derivation of Key 1 (Section 3.4). Finally, Shcherbak and Makukov work extensively with the euplotid code variant (in which TGA is reassigned to Cys), which they take as a symmetrized counterpart of the standard code; Panov and Filatov treat this as a switchable alternative. The present work focuses on the standard code, and compares its arithmetic structure with the 26 alternative NCBI translation tables on equal footing (Sections 4.1, 4.1).

We now describe how the divisibility findings of these prior works project onto the groups defined in the present analysis. Shcherbak and Makukov, counting one molecule per distinct coding role within a family, observed that the sum of amino acid nucleon counts over Octet I is divisible by 37, with $T(\text{Octet I}) = 925 = 25 \cdot 37$, and that the same holds for Octet II under the assignment of zero to stop codons, with $T(\text{Octet II}) = 2220 = 60 \cdot 37$. Panov and Filatov, having introduced the per-codon counting rule, obtained the corresponding totals under codon-level counting: $T(\text{Octet I}) = 3700 = 100 \cdot 37$ and $T(\text{Octet II}) = 4218 = 114 \cdot 37$ (or $4440 = 120 \cdot 37$ under their alternative assignment of 74 to stop codons).

The relation between the per-coding-role total and the per-codon total is direct for Octet I, which is fully four-fold degenerate: $3700 = 4 \cdot 925$. For Octet II, whose families contain different numbers of distinct coding roles, the per-codon total 4218 is not a multiple of the per-coding-role total 2220 and is an independent fact. The per-coding-role total 2220 itself coincides with $T(\text{Keto} \cap \text{Octet II}) = T(\text{Strong} \cap \text{Octet II})$ in our analysis: the universal pyrimidine synonymy within Octet II (Section 2.4) implies that the sixteen codons of Keto (or of Strong) intersected with Octet II contain each coding role exactly once, so their per-codon total equals the per-coding-role total.

From these primary facts, the divisibility by 37 of several further groups in our systematics follows trivially. The total over the entire code is the sum of the two octet totals; the sixteen-codon and eight-codon sub-groups of Octet I are mutually identical by four-fold degeneracy; and the four chemical-axis groups Keto, Strong, Amino, Weak each decompose into an Octet I sixteen-codon sub-group and an Octet II sixteen-codon sub-group, of which the first is automatically divisible by 37 and the second reduces to a separate question addressed in the present work.

4.5 Limitations and outlook

The statistical significance of the observed arithmetic regularities is assessed in Section 4.3 under one explicit null — the random-code permutation, which fixes the code's degeneracy architecture and the amino-acid nucleon counts and randomizes only the assignment of amino acids to codon families. Under that null the headline divisibility count and the continuous concentration over the full sense pool are highly atypical. The narrower S3 test, restricted to nested groups that contain no service codon at all, is only weakly atypical ($p_{\text{MC}} \approx 0.025$) and mainly reflects copies of the Octet I total. The choice of null is not unique: alternative generative models, for example randomizing the block sizes themselves or the nucleon counts, would probe different aspects of the regularity and are left for future work, as is any account of a mechanism that could produce it.

The parametrization of service codons in the present work is restricted to Key 0 and Key 1; additional parametric assignments are tabulated in `key_parameters.csv` and in the accompanying `visualization.html` for illustration of other choices.

A further direction concerns the group-theoretic structure of the codon table. Following Panov and Filatov [5], to each of the 24 orderings of the four bases on the axes of the codon matrix we associate its *calligram*: the geometric pattern formed by the eight homogeneous codon families of Octet I on the 4×4 matrix. Connectedness of an octet refers to the connectedness of this pattern under 4-adjacency of matrix cells. Among the 24 calligrams, Panov and Filatov [5] identified four — CTGA, CGTA, ATGC, and AGTC — in which each of the Rumer octets forms a connected region, and showed that these four form a closed set under the transformations $R = (\text{T} \leftrightarrow \text{G}, \text{C} \leftrightarrow \text{A})$, $R_1 = (\text{T} \leftrightarrow \text{G})$, and $R_2 = (\text{C} \leftrightarrow \text{A})$. These three transformations together with the identity form the Klein four-group V_4, and the action of V_4 extends naturally to the full set of 24 calligrams. This action of V_4 on the full set of 24 calligrams is free, partitioning it into six orbits of four calligrams each. Exactly one of these six orbits consists of connected calligrams, accounting for the four calligrams above; the remaining five orbits contain no connected calligram. This connected orbit furthermore contains the canonical chemical ordering C, G, T, A of Rumer [7] (Section 2.2); the correspondence between this chemical ordering and the geometric property of connectedness is non-automatic, since only two of the eight chemically alternating orderings of the four bases yield connected octets. The search for further structural invariants distinguishing the orbits — and for connections between the group-theoretic orbits and the arithmetic systematics of the present work — is left to future investigation.

5 Conclusion

The standard genetic code exhibits arithmetic regularities visible at several levels of parametric specificity. A first layer is independent of any choice of nucleon contributions for the service codons, and it carries the central finding of this work: summed over the sixty sense codons, the neutron pool is an exact multiple of the modulus, $N(\mathcal{C}_{\text{SE}}) = 3589 = 97 \cdot 37$, while the proton pool falls one unit short of an odd multiple, $P(\mathcal{C}_{\mathrm{SE}}) \equiv -1 \pmod{37}$—a property of the proton/neutron split that is invisible at the level of the total count T. Two further parametrization-independent facts sit alongside it. The eight families of Octet I carry $T(\text{Octet I}) = 3700 = 100 \cdot 37$, with $P(\text{Octet I}) = 2000$ and $N(\text{Octet I}) = 1700$, the latter combining with the Octet II sense pool as $1700 + 1889 = 3589$; and the entire Keto/Amino asymmetry of the code localizes, after exclusion of the service codons, to the single amino-acid pair Trp–Ile, with $|\delta N| = 37$ and $|\delta P| = 36$. The universal pyrimidine synonymy within Octet II produces identities between the twin sixteen-codon groups Keto/Strong and Amino/Weak within Octet II.

A second layer arises under the naive reading, where the proton and neutron counts P and N are analyzed separately. The total nucleon count T is divisible by 37 for the four chemical-axis groups and for Octet II. For the Amino and Weak groups, P and N are individually divisible by 37, aligning these groups with the lattice of step 37 along both axes.

A third layer introduces the service-codon assignment. There is a unique minimal non-negative integer assignment, Key 1 (Section 3.4), under which the parametrization-independent structure extends to the full 64-codon table. Key 1 is not a foundation for the findings above but a device that renders them on the complete table: the same sense-pool divisibility, the same localized pair, and the same proton parity that hold with the service codons unassigned are exactly what make the assignment solvable in small integers. The neutron balance across the Keto/Amino axis reproduces the Trp–Ile difference $|\delta N| = 37$; the global neutron deficit $T(\text{Octet I}) - N(\mathcal{C}_{\mathrm{SE}}) = 111 = 3 \cdot 37$ closes because it is a common multiple of the number of stops and the modulus; the even-proton parity theorem forbids $|\delta P| = 37$ and bars the sense-pool proton pool from the odd multiples of the modulus, so that it sits one unit below—by a margin that the specific nucleon counts make exactly one—which the start codon's minimal contribution $p_{\mathrm{start}} = 1$ restores; and $\gcd(4, 37) = 1$ leaves a unique minimal solution. None of these is a parametric assumption; each is an empirical property of the standard code established at the parametrization-independent level, and Key 1 exists as the minimal extension precisely because they hold simultaneously.

Under the derived parametrization, the arithmetic regularities organize into a connected family: alignment of the four chemical-axis groups along the axes T, P, N, and Δ (Section 3.5.1), macroscopic identities including $N(\text{All}) = T(\text{Octet I}) = 3700$ and $P(\text{All}) = T(\text{Octet II}) = 4292$ (Section 3.5.2), the uniform invariant $\Delta = 296$ across all six 32-codon groups (Section 3.5.4), and divisibility patterns by 37 and 999 at multiple levels of partitioning (Section 3.5.3). The comparative analysis across the 27 NCBI translation tables (Section 4.1) shows that the simultaneous holding of all structural preconditions enumerated above is a property of the standard code alone.

Two features of this structure constrain how it may be read. First, the regularities reside in group sums rather than in individual molecules: no single canonical amino acid has a nucleon count divisible by 37, and in groups of variable coding degeneracy, such as Amino, the divisibility holds only when codons are counted with their multiplicity and fails when each amino acid is counted once. The nucleon composition and the degeneracy of coding are therefore not independent quantities within this structure: the regularity is a joint property of how nucleon mass is distributed and how many codons each amino acid receives. Second, the entire first layer above, including the values over Octet I and the twin-group identities, is established with the service codons carrying no nucleon values at all; the derived parametrization of the service codons enters only at the later layers and does not affect the parameter-independent results. The assigned values are thus a device for exhibiting the arithmetic of the complete table, not a foundation on which the principal findings rest.

The specificity established by the comparison across natural codes is complemented by deterministic controls and by a Monte Carlo analysis. The controls show that divisibilities concentrate in third-position groupings and that 37 stands out among the tested prime moduli. Under an explicit random-code null that fixes the degeneracy architecture and permutes the twenty real paired proton and neutron profiles over the twenty blocks, two dependent sense-pool summaries place the standard code in the far tail: the discrete S2 count gives $p_{\mathrm{MC}} = 4.0 \times 10^{-5}$, and the S4sense mean distance gives $p_{\mathrm{MC}} = 2.74 \times 10^{-4}$. The narrower S3 test, restricted to groups that contain no service codon at all, is only weakly atypical ($p_{\mathrm{MC}} \approx 0.025$) and mainly reflects dependent copies of the divisibility of T(Octet I). The concentration persists under Key 0 ($p_{\mathrm{MC}} \approx 0.0028$), whereas the symmetric test of the Key 1 derivation gives moderate atypicality ($p_{\mathrm{MC}} \approx 0.021$). These probabilities refer to different, dependent summaries and are not combined by multiplication. This is improbability under one combinatorial null, distinct from the specificity among the 27 natural codes, and it speaks to no biological mechanism; alternative null models are left for future work.

The methodological choice underlying the present work is to restrict attention to parameters whose definition is unambiguous — the total nucleon count T, its decomposition into protons P and neutrons N, and the difference $\Delta = P - N$ — and to a single grouping system based on the Rumer octets and the third-position chemical axes. The interpretation of the arithmetic facts collected here is left open.

References

[1] Michael Berglund and Michael E. Wieser. Isotopic compositions of the elements 2009 (IUPAC technical report). *Pure and Applied Chemistry*, 83(2):397–410, 2011. doi: 10.1351/PAC-REP-10-06-02.

[2] Stephen J. Freeland and Laurence D. Hurst. The genetic code is one in a million. *Journal of Molecular Evolution*, 47(3):238–248, 1998. doi: 10.1007/PL00006381.

[3] Sunghwan Kim, Jie Chen, Tiejun Cheng, Asta Gindulyte, Jia He, Siqian He, Qingliang Li, Benjamin A. Shoemaker, Paul A. Thiessen, Bo Yu, Leonid Zaslavsky, Jian Zhang, and Evan E. Bolton. PubChem 2023 update. *Nucleic Acids Research*, 51(D1):D1373–D1380, 2023. doi: 10.1093/nar/gkac956.

[4] National Center for Biotechnology Information. The genetic codes, 2024. URL `https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi`. Last updated 23 September 2024; accessed 22 July 2026.

[5] Alexander D. Panov and Felix P. Filatov. Are the strange information structures of the genetic code an accident or an artifact? In *Evolution: Environmental, Demographic, and Political Risks 2024*, pages 23–68. Uchitel Publishing House, 2024. doi: 10.30884/978-5-7057-6399-3_03.

[6] Yu. B. Rumer. Codon systematization in the genetic code. *Doklady Akademii Nauk SSSR*, 167: 1393–1394, 1966. URL `https://pubmed.ncbi.nlm.nih.gov/5997290/`. In Russian. PMID: 5997290.

[7] Yu. B. Rumer. Systematization of codons in the genetic code. *Doklady Akademii Nauk SSSR*, 183:225–226, 1968. URL `https://www.prometeus.nsc.ru/science/schools/rumer/works/1968.ssi`. In Russian.

[8] Vladimir I. Shcherbak and Maxim A. Makukov. The “Wow! signal” of the terrestrial genetic code. *Icarus*, 224(1):228–242, 2013. doi: 10.1016/j.icarus.2013.02.017.